

Development Team: Group 11

Wicked Realty (Group 9)

Requirements Documentation

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Revision History

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1. Overview

Wicked Realty is commissioning the development of a residential real estate Fair Value Estimate (FVE) engine to modernize and automate its property valuation process. Currently, the firm relies on manual calculations performed by realtors to determine market values, a process that is both time-consuming and error-prone. This new FVE engine will replace subjective “gut feel” valuations with standardized, defensible reports through a Python-based command line interface (CLI). By integrating features such as Geospatial Location Weighting and Adjustment Logic for specific home features, the FVE engine aims to provide consistent and rapid Comparative Market Analysis (CMA) reports. Ultimately, the FVE engine will serve as a single source of truth for Wicked Realty and support high-volume market valuation analysis to allow realtors and agents to focus on customer relations.

2. Business Requirements

a. Background

Wicked Realty lacks an automated system for generating FVE of residential properties in Norman, forcing realtors to perform all CMA calculations manually. This manual process relies

on metrics such as the number of bedrooms, bathrooms, square footage, and a number of other relevant factors. However, since these CMAs are performed by individuals, the FVEs generated are highly susceptible to “gut feel” biases and human error. Wicked Reality wants an algorithm to rapidly compute the FVE of properties from a CMA perspective more efficiently and effectively serve their customers by providing a plethora of data driven FVE.

b. Business Opportunities

Wicked Realty’s current bottleneck in getting an estimate for potential home buyers is doing the manual calculations on what the FVE of a home is. The FVE engine will algorithmically compute consistent FVEs, heavily reducing human workload and dramatically speeding up the CMA process. This also reduces possible errors caused by realtors overlooking or miscalculating figures. Overall, this product will increase organizational efficiency, reduce errors, and provide a more consistent output for the realtors and Wicked Realty’s customers.

c. Business Objectives

- The average time to calculate the FVE on a residential property should be reduced from 1 hour to under 1 minute immediately after implementing the system.
- The average time to calculate multiple FVEs, to a maximum of 100, should be reduced from multiple business days to under 5 minutes immediately after implementation of the system.
- The volume of property appraisals handled by Wicked Realty per week should increase by 40% over the following month after implementing the system.

d. Success Metrics

- ROI is calculated as

$$ROI = \frac{WagesSaved - FVECost}{FVECost}$$

where the wages saved are

$$WagesSaved = TimeSaved \cdot EmployeeHourly \cdot NumEmployees$$

and FVE Cost is the cost of developing and maintaining the FVE engine, which will be close to zero. Ideally, a high ROI of over 300% should be achieved.

- Accuracy delta is measured as the average accuracy of the FVE minus the average accuracy of the realtors, where accuracy is measured as

$$\text{Accuracy} = 1 - \left| \frac{\text{Estimate} - \text{Actual}}{\text{Actual}} \right|$$

The accuracy delta should be at least 2%.

- DOM is calculated as

$$\text{Average DOM} = \frac{\sum \text{Days on Market}}{n}$$

where n is the number of properties. As the algorithm will run faster and be more accurate than realtors, the average DOM should drop by at least 10%.

e. Product Vision Statement

For Wicked Realty realtors and agents who need to rapidly generate accurate and consistent FVEs for residential properties without relying on slow, error-prone manual calculations, the FVE engine is a centralized, Python-based algorithmic tool that will provide standard Python-based, feasible, and transparent valuations by automating CMA through geospatial weighting and adjustment logic. Unlike Wicked Realty's current manual processes that consume significant man-hours and rely on subjective "gut feel", the FVE engine eliminates the human element from core computations, maximizing operational efficiency by reducing CMA computation time from one hour to under 5 minutes and allowing realtors to focus on customer relations and sales.

3. Scope and Limitations

a. Major Features

- FVE calculation for a single property
- FVE calculation for a batch of properties of up to 100 properties
- Comprehensive CMA Input Engine that allows for custom inputs
- Geospatial Location Weighting for neighborhood specific premiums
- Standardized Adjustment Logic that allows for custom weights

b. Project Scope

The scope of this project involves the development of an FVE engine to provide a standardized, defensible, and transparent framework for property valuation, moving away from subjective "gut feel" estimates. By automating the CMA process, the tool aims to eliminate the human element from core computation processes, thereby mitigating calculation errors and implementing a consistent and scalable computation operation across the entire agency. This tool strategically aligns with Wicked Realty's long-term goals of maximizing operational efficiency; by offloading data-heavy analytics to computers, realtors are freed to focus on

customer relations and sales. This project directly supports the corporate goal of digital transformation, positioning Wicked Realty as a data-driven leader in local markets. The software will serve as a single source of truth for all residential listings, enabling inter-agent and inter-departmental collaboration while lowering the barrier to entry for realtors to be effective in this field of work.

c. Limitations and Exclusions

To maintain project focus and ensure timely delivery, certain functionalities have been explicitly excluded from the current scope. Firstly, the FVE engine will provide a snapshot valuation only and will not include predictive forecasting or market trend modeling; the tool is designed to deliver a property's current FVE rather than future appreciation. Secondly, the system will be developed to FVE generation of residential property listings. The algorithmic logic will be calibrated for housing metrics and cannot accurately support commercial real estate, which is affected by starkly different factors, making it a uniquely different product.

Thirdly, automated data scraping is excluded from this release. While integration with large third-party platforms such as Zillow, Homes.com, Realtor.com, and other property listing marketplace platforms is a potential future iteration, the current version will require all input parameters to be handled by the firm's staff. The development will provide an initial dataset alongside the pre-beta release to show working proof of concept.

These exclusions are intentional to prioritize the core FVE engine accuracy and tailor the system's overall ease of use for Wicked Realty's staff. By defining these boundaries, the team can ensure timely delivery of a stable, reliable tool that meets Wicked Realty's immediate internal requirements whilst balancing the cost, maintenance, and integration complexities of external dependencies.

4. Context Description

a. User Classes and Characteristics

The product has 3 main user classes to be implemented. The primary user class is the realtors. Realtors should be the main users of the product, possibly using it every day. Realtors are expected to have high knowledge of the parameters and should be able to identify values for each parameter. However, there is no expectation that the realtors should have any technical knowledge. Clear labels must be included on every interface, using only terminology that realtors should be comfortable with. Important actions should be highlighted to grab attention.

New users should be able to create an output within 5 minutes. Error messages should describe the problem and what is necessary for the user to fix it.

The administrative managers (managers) are a secondary user class. They are expected to have no technical knowledge. Managers' primary use case is for reading high-level reports and managing the realtors. Managers are responsible for managing realtors' access to the product. This includes providing and removing access to the realtors.

I.T. administrators and software developers are a secondary user class responsible for maintaining the product post launch. They should be responsible for bug fixes and updating the algorithm as the market shifts. They are expected to have some technical expertise. The UI for the IT/Dev user class should have clear labels and descriptive error messages.

b. Operating Environment

The software will operate within Wicked Realty's internal computer network. It will be a simple, text-based program (known as a command line interface) built using Python. Employees just need a standard computer that can open text-based programs and basic charts. The program must work smoothly alongside spreadsheet software like Microsoft Excel and Google Sheets, so users can easily load and save their housing data. It might also need to connect to the company's secure databases to store this information safely.

c. Design and Implementation Constraints

The development of the FVE engine is governed by several strict technical and operational constraints designed to ensure immediate usability and long-term sustainability. The primary interface deliverable must be a CLI built using Python. To maintain data security and operational independence, the system will be built to support locally hosted housing data and will be restricted from using external APIs, although this is possible for future iterations. The engine must be maintainable, allowing non-technical managers, realtors, and agents to adjust factor weights, such as neighborhood premiums or feature modifiers, through external configuration rather than requiring code changes, ensuring the FVE engine remains adaptable and current to market shifts.

d. Assumptions and Dependencies

This project operates under a few assumptions based on data, users, and maintenance. First, it is assumed that information about housing data will be up to date, sufficient, accurate, and

stored in a CSV format. Furthermore, we assume that the housing data will be complete, consistent, and represent the area where the house is located accurately. It is assumed that end users will have access to computers capable of running console applications and basic data visualization tools. As for post development lifecycle, the project assumes that Wicked Realty will allocate necessary personnel and resources to maintain and upkeep the system and data updates. Finally, the development of the system is going to be heavily dependent on using the Python programming language, and scikit learn machine learning library to build the predictive model, alongside the strict dependency that all historical data must be queried from a CSV file rather than from an external API.

e. Glossary of Terms

Fair Value Estimate (FVE)

Fair Value Estimate refers to a standardized, market adjusted valuation of real estate listings after variables such as housing conditions, property features, and other relevant variables are taken into account.

Comparative Market Analysis (CMA)

A method used by real estate professionals to estimate the value of a property by evaluating similar listings within an area to determine a fair and competitive market price.

Geospatial Location Weighting

A mathematical adjustment applied to the valuation algorithm that accounts for the geographical desirability of specific neighborhoods and zones.

Comma separated List (CSV)

A file format for storing data in a tabular format where values are separated by commas and organized into rows and columns

Command line (CLI)

A user interface used to run programs

Return of Investment (ROI)

A performance measure used to evaluate the efficiency of an investment. This is calculated by using the $(\text{annual return} / \text{total investment}) \times 100$. In the context of real estate, we are aiming for 10% or higher to be classified as 'strong'

Application programming interface (API)

A set of rules and protocols that allow different software applications to communicate and exchange data with each other using functions such as "GET" and "POST."

Accuracy Delta

The numerical difference between the fair value price predicted by the algorithm and the actual final sale price of the property; used as a primary metric to evaluate the model's performance.

Days on market (DOM)

total number of days from the date a property is first listed in the system to the day of its sale if applicable

User interface (UI)

Regression Model

A type of machine learning algorithm used to predict a continuous numerical value

IT/Dev

personnel responsible for the technical deployment, hosting environment, and long-term maintenance

f. References

Real Estate Price Listing Websites

- Source: Zillow and Apartments.com.
- Location: zillow.com and apartments.com.
- Purpose: External web addresses referenced for researching the price, square footage, and attributes of homes in the assumed area.

Wicked Realty Housing Price Predictions RPF

- Authors: Wicked Realty Project Team 9 (Steven Tran, Sean Ropp, Trevor Bean, Caleb Furley, Alex Requena, Abhinav Pillai).
- Version: 1.0
- Date: January 28, 2026.
- Purpose: The primary source document establishing the project's background, objectives, and constraints.

5. System Features

System Feature 1: Docker Image

Priority: MEDIUM

Description: The Docker Image feature packages the FVE engine along with all necessary dependencies and environment configuration into a single, standardized container. This ensures that the application runs identically across different workstations and operating systems – mitigating the need for onerous installations and configuration – simplifying deployment and maintenance.

Functional Requirements: These are the software capabilities that must be present in order for the user to carry out the services provided by the feature, or to execute related use case(s).

I) DOC-REQ-1: Containerized Runtime Environment

a. Responding to Anticipated Error Conditions

i. Missing Docker Daemon

If the user attempts to run the initialization script before the Docker engine is installed or running on the host machine, the setup script shall gracefully exit and echo “Docker not found” error.

ii. Insufficient Resources

If the host machine does not meet the minimum memory allocations required by the container, the container shall gracefully fail on startup and log an “Insufficient Memory Allocation” error.

b. Backward Traceability

i. Source

- Client Meeting – 02/09/2026

ii. Rational

After discussion with the client about operating system capability constraints, the development team determined that an isolated container prevents Environmental Inconsistency errors. Running the CLI application also allows for easy version control without managing complex environments across the firm.

c. Forward Traceability

User Acceptance Testing: After deploying the image onto a clean test host computer, if the CLI application runs successfully, this requirement is met and verified.

II) DOC-REQ-2: Local Data Volume Mounting

a. Responding to Anticipated Error Conditions

i. Volume Mount Permission Denied

If the container attempts to access a protected local directory or lacks read/write permission for target folders, the CLI shall echo a storage access error and exit execution with exit code in accordance with CLI-REQ-1.

b. Backward Traceability

i. Source

- RFP Section 6.0
- RFP Section 7.0

ii. Rationale

The system will use local CSV data for this current iteration of application since cloud integrations are complex and will delay the development cycle of the application. However, it is possible to stub out potential external API and cloud integrations for future development.

c. Forward Traceability

Integration Testing can verify that CSV files placed in predetermined local host directories can be successfully ingested by the containerized CLI application, and that resulting reports can be saved to predetermined local host directories.

Use Cases:

- DOC-REQ-1: The system shall successfully instantiate the Docker container and present the application CLI prompt to the user in no more than THIRTY (30) seconds of executing the run command.
- DOC-REQ-2: Upon composing the Docker container, the system shall initialize the container and establish a read/write connection to the configured local folders within TEN (10) seconds.

Use Case Flow:

1. User downloads the Wicked Realty FVE engine Docker image to their local workstation.
2. User runs the initialization command mapped to their local CSV storage directory, triggering DOC-REQ-2.
3. System triggers DOC-REQ-1 which unpacks necessary environments and dependencies without touching the host OS registry.
4. System presents the active CLI application to the user.

User Stories:

User 1

Narrative:

- As an I.T. Administrator,
- I want to deploy the CLI application as a pre-packaged Docker image,
- so that I will not have to manually install all environments, versions, and dependencies on every single workstation.

Acceptance:

- Given a new workstation with Docker engine installed,
- When I pull Wicked Realty FVE engine Docker image and execute the docker run command,
- Then the CLI application will be fully operational and ready for use without any further setup.

User 2

Narrative:

- As a Real Estate Agent,
- I want to run the application easily on my local computer with my local files,
- so that I will not have to go through extensive setup before using the application to generate reports.

Acceptance:

- Given CSV files in my local host folders,
- When I launch the application and run designated commands,
- Then the application can read my local CSVs and save reports onto my local folders.

System Feature 2: Command Line Interface Application

Priority: HIGH

Description: Provides a lightweight, text-based interface within a terminal for users.

Functional Requirements

I) CLI-REQ-1: Command Execution & Parsing

a. Responding to Anticipated Error Conditions

i. Missing Dependencies

If commands require external files that are not found, the CLI shall provide the specific path, file name, and file extension of where and what was expected.

ii. Program Execution Error

If a program exits with any code other than a success code (CODE:0), the CLI shall:

- **Terminate Execution:** Immediately halt the current process to prevent cascading errors and further data corruption.
- **Return Standard Error:** Output a concise, human-readable error message to `stderr` which highlights exactly which step failed.
- **Log the Exit Code:** Echo the exit code with brief description to assist troubleshooting. I.e `Exit Code 124: Command Time Out`.

b. Backward Traceability (rationale for the requirements and the source – RFP and which section in it, client meeting and which notes from that meeting, etc.)

c. Forward Traceability

i. Verification

1. User Acceptance Testing (UAT) via terminal execution using Blackbox testing methodologies.
2. Automated unit tests for command parser accuracy.

II) CLI-REQ-2: Help Documentation

a. Responding to Anticipated Error Conditions

i. Invalid Syntax

If a command is entered incorrectly, the CLI shall echo the invalid syntax back to the user with a suggestion for the syntax, or a master help list that shows all available commands.

b. Backward Traceability: Traces to section 4a (User classes and characteristics), ensuring that realtors with no technical backgrounds have clear labels, descriptive error messages, and immediate access to command syntax rules

c. Forward Traceability: Verified by UAT where users intentionally input malformed commands and assert that the system successfully returns the help dialog rather than crashing.

Use Cases:

- CLI-REQ-1: i.e. The CLI shall parse user commands and initiate the backend python processes within ONE (1) second of the user pressing the return key
- CLI-REQ-2: The user types 'fve-engine --help' and the system responds with a formatted ascii menu containing a list of valid commands, required arguments, and usage examples

Use Case Flow:

- The user inputs a terminal command (e.g., fve-engine calculate --batch properties.csv).

- CLI-REQ-1 parses the command string to identify core commands, flags, and arguments.
 - o If syntax is invalid, CLI-REQ-2 halts execution and outputs the help menu.
 - o If syntax is valid, CLI-REQ-1 executes the requested process.
- Upon termination, the CLI captures the exit code, printing success to stdout or routing errors to stderr.

User Stories:

User 1

Narrative:

- As a Real Estate Agent,
- I want to be able to pull up a list of instructions directly in the terminal,
- so that I can quickly figure out how to format my command without having to search for a separate instruction manual.

Acceptance:

- Given the user is in the terminal and unsure of the exact command syntax,
- When the user types five-engine --help,
- Then the CLI immediately outputs a clean, readable list of all commands, what they do, and an example of how to type them.

User 2

Narrative:

- As a Software Engineer,
- I want the CLI to return specific exit codes and standard errors when a command fails,
- so that I can quickly debug issues or write automated scripts that know exactly why a batch job was terminated.

Acceptance:

- Given a user attempt to run a calculation but provides a file path that does not exist,
- When the system attempts to parse the command and the file,
- Then the CLI stops execution and returns a clear error along with the relevant non-zero exit code.

System Feature 3: Data Sanitation and Preprocessing

Priority: HIGH

Description: The Data Sanitation and Preprocessing module is responsible for ingesting raw property data, validating structure and content, normalizing inconsistent values, and preparing a clean standardized dataset to later be processed by the adjustment model. This feature acts as a data quality gate for the entire Wicked realty pipeline.

Functional Requirements

I) DSP-REQ-1: input validation and schema enforcement

The system shall validate all CSV data against a previously defined schema, ensuring all fields (bedrooms, bathrooms, etc) exist and conform to expected types

a. Responding to Anticipated Error Conditions

i. Missing Required Fields

If the CSV is missing required columns or if CRITICAL cells are empty, the system will skip the affected row and log “missing data” error specifying that the exact row and column to stderr and continue processing the rest of the batch.

ii. Incorrect Data Types

If a field contains invalid data types like a string “five” instead of an integer 5 in a numeric column, the system will attempt to cast it to correct the typing mismatch. If the casting is to fail, the row is rejected and an “invalid Data Type” error is logged indicating the specific row and field where it occurred.

b. Backward Traceability: Traces back to Business Requirements Section 2a (mitigating manual calculation error) and section 4d (assumptions that data will be stored and retrieved in a consistent CSV format)

c. Forward Traceability: Verified through automated unit testing where mock CSVs with known structural flaws (missing columns, strings in integer fields) are fed to the parser, asserting that the engine correctly drops invalid rows and generates accurate logs without crashing the application.

II) DSP-REQ-2: Data Normalization and Cleaning

The system shall normalize all text-based inputs and handle standardizing addresses and string values before passing the data to the analytical engine to ensure accurate categorical matching.

a. Responding to Anticipated Error Conditions

i. Out of Bounds / Anomaly Values: If numeric values exceed logical real estate boundaries such as a house with 0 sqft or 500 bedrooms, the

system will flag the entry as "Anomalous Data" and exclude it from the analysis to prevent mathematical skewing in the FVE output.

- b. Backward Traceability: Traces to section 3a (Comprehensive CMA Input Engine) to ensure custom inputs are standardized before geospatial and adjustment processing occurs.
- c. Forward Traceability: User Acceptance Testing where agents input purposefully messy data like erratic capitalization, trailing whitespaces, etc... and verify that the system successfully parses the file and outputs an FVE identical to one generated from a perfectly formatted file.

Use Cases:

- DSP-REQ-1: The system shall parse a standard batch CSV file (up to 100 properties) and identify all schema violations, dropping invalid rows in under TWO (2) seconds.
- DSP-REQ-2: The system shall automatically apply trimming, casing standardizations, and basic data casting to all relevant feature inputs upon CSV ingestion.

Use Case Flow:

- 1) The user executes the CLI command to trigger FVE calculation passing the local path to a target CSV file
- 2) The system triggers DSP-REQ-1 to open the file and verify the presence of all mandatory columns based on the internal schema.
- 3) The system triggers DSP-REQ-2 to iterate through the rows, trimming whitespace, standardizing text casing, and checking out of Bounds Values.
- 4) If the system encounters missing required Fields or incorrect data types, the invalid row is dropped from memory, and the error is appended to the log.
- 5) The system outputs a brief sanitation summary to the CLI
- 6) The finalized, clean dataset is passed to the core FVE engine for CMA calculations.

User Stories:

User 1

Narrative:

- As an IT admin,
- I want the system to explicitly log which rows and columns failed to validate,
- so that i can easily inform realtors which specific cells they need to fix in their CSV without having to hunt down the corrupt data.

Acceptance:

- Given a CSV file with a missing square footage data on row 12,
- When the user runs the batch calculation command,
- Then the system skips row 12, processes the rest of the file and outputs an error to the terminal reading "Row 12: missing required field 'square footage'".

User 2

Narrative:

- As a Real Estate Agent,
- I want application to automatically fix minor formatting issues like extra spaces or inconsistent capitalization,
- so that I do not have to spend excessive time perfectly formatting every cell before running my reports.

Acceptance:

- Given a CSV file where a neighborhood parameter is entered with bad formatting such as "Chautauqua ",
- When the system ingests the data via the CLI,
- Then the preprocessing module silently normalizes the string to "Chautauqua" and processes the FVE successfully without interrupting the user.

System Feature 4: Fair Value Estimate (FVE) Engine

Priority: HIGH

Description: The FVE engine will serve as the core analytical backbone of Wicked Realty, designed to automate and standardize the creation of CMA reports. This engine will also account for geospatial location weighting which dictates market premiums and provide a more precise property valuation.

Functional Requirements

- I) FVE-REQ-1: Geospatial Weighting Calculation
 - a. Responding to Anticipated Error Conditions
 - i. Missing Coordinate Data

If coordinate data for CMA is missing or malformed, the system shall fall back to city-level weighting metrics and append the degraded geospatial precision used to the calculation log.
 - ii. Zero Comparables Found

If no comparative properties are found within the given geospatial radius, the FVE engine will skip the CMA calculations for affected properties and

flag them with “Insufficient Market Data” error to indicate that manual calculations are needed for a CMA.

- b. Backward Traceability (rationale for the requirements and the source – RFP and which section in it, client meeting and which notes from that meeting, etc.)
- c. Forward Traceability (how requirements can be verified by users)

Comparing the FVE CMA calculations against a verified, known coordinate dataset and asserting a difference of no more than TEN (10) percent margin of error.

II) FVE-REQ-2: CMA Report Generation

a. Responding to Anticipated Error Conditions

i. File Write or Permission Failure

If the system lacks the necessary directory permission or storage space to save the generated CSV report, it shall halt the export process and exit with the relevant Exit Code in accordance with CLI-REQ-1.

b. Backward Traceability (rationale for the requirements and the source – RFP and which section in it, client meeting and which notes from that meeting, etc.)

i. Source

- RFP Section 1.0
- RFP Section 3.0
- RFP Section 7.0
- Client Meeting – 02/09/2026

ii. Rationale

The RFP explicitly stated that current processes for property calculations were done by hand, which is time-consuming. The core objective to build a CMA calculator was to automate these pricing tasks whilst reducing expensive human error in the calculation process. Whilst the client initially requested an AI pipeline, further discussion revealed that an easy to maintain, light weight, and cost-effective solution was heavily favored which led to a pivot away from AI workflows.

c. Forward Traceability

UAT where real estate agents, IT administrators, and other actors trigger the CMA report generation for a target property. Testers will download the resulting CSV and verify that the data is coherent in spreadsheet applications like Google Sheets or Microsoft Excel. Automated integration tests will also verify that CSV file report is generated within acceptable parameters.

Use Cases:

- FVE-REQ-1: The FVE engine shall calculate the baseline FVE and apply geospatial location weighting based on comparative sales within a defined radius, completing calculations in no more than FIVE (5) seconds for each property.
- FVE-REQ-2: Upon a successful completion of the CMA calculations, the system shall automatically trigger a CSV file export to a configurable, predetermined directory before echoing a success message in the CLI.

Use Case Flow:

1. User inputs parameters of a target property and gives the CLI command to trigger a CMA calculation.
2. System preprocesses the input parameters and triggers FVE-REQ-1 to apply geospatial weight coefficients based on the property's location to compute a baseline FVE.
3. The system will then trigger FVE-REQ-2 and export the baseline FVE as a CMA report in the form of a CSV file in the specified directory.
4. The system will notify the user of successful completion of the CMA report in the CLI and log the successful export in the system audit trail.

User Stories:

User 1

Narrative:

- As a Real Estate Agent,
- I want to a simple and automatic process that calculates a property's precise value accounting for location-specific market premiums,
- so that I can provide a defensible, data-backed estimate to my clients while lowering the total manhours required for a valuation.

Acceptance:

- Given a target property and a validated CSV containing all comparable property data with valid geographic coordinates,
- When the actor executes the FVE calculation command in the CLI application,
- Then the system must calculate the baseline FVE by applying the geospatial weighting and update the final valuation within the generated report in no more than FIVE (5) seconds.

User 2

Narrative:

- As an Operations Manager,

- I want to export the completed CMA data into a CSV file,
- so that I can easily import the data into Excel for bulk portfolio analysis.

Acceptance:

- Given the FVE engine has successfully completed FVE reports for one or multiple properties,
- When the calculation workflow concludes,
- Then the system must automatically fetch a copy of the report into a configured output directory accessible to management staff as a CSV file that can be opened in spreadsheet applications like Google Sheets or Microsoft Excel.

User 3

Narrative:

- As a I.T. Administrator,
- I want to access detailed error logs whenever there is a process or execution failure in the FVE engine pipeline,
- so that I can rapidly troubleshoot and pinpoint failure points to fix the application.

Acceptance:

- Given the FVE engine encounters a terminal issue during execution such as directory permission failure,
- When the system halts the current calculation process,
- Then the system must echo the relevant Exit Code to the CLI and append a detailed error message including the timestamp, error type, and affected data properties to the system audit log.

System Feature 5: Adjustment Logic Model

Priority: MEDIUM

Description: The custom weight adjustment logic model is a system that puts weights on different aspects of the property to adjust the price according to a customer's needs/wants. For example, if a customer is willing to pay more for a pool or four bedrooms, the model will be able to better reflect that by increasing the price on properties with this parameter. This model will also be able to lower the price on properties with aspects the customer does not want.

- I) ADJ-REQ-1: Client Adjustment Profile Consumption
 - a. Responding to Anticipated Error Conditions
 - i. Incorrectly formatted Client Profile

If the CSV is missing or improperly formatted, the system shall not proceed to calculation, exit with the relevant Exit Code in accordance with ADJ-REQ-1.

ii. Weight value out of bounds

If the input weight value lowers the estimate of the home below 0, the system shall exit with the relevant Exit Code in accordance with ADJ-REQ-1.

b. Backward Traceability: This system will provide a method for home buyers to list parameter for a home based on its attributes, providing this information to the secondary FVE calculation for decision-making purposes to potential home buyers. This requirement is listed in the RFP under section 5.0, primary stakeholder 2, potential buyers and researchers.

c. Forward Traceability: Multiple unit tests containing valid and invalid CSVs will be created and ran into the system to ensure CSV are the correct format and no values are out of bounds.

II) ADJ-REQ-2: Client-Adjusted FVE Calculation

a. Responding to Anticipated Error Conditions

i. Unmatched Adjustment Parameter

If the Client Profile Adjustment CSV specifies weighting for certain features (i.e. Cul De Sac) that does not exist within the property's data, the system shall safely ignore that specific weight in the adjustment calculation and proceed with all other matches parameters. The system will then log which feature was not matched in the CMA adjusted report.

ii. Missing Baseline FVE Dependency

If the secondary adjustment calculation is triggered, but the baseline FVE is not yet calculated or unable to be found, the system shall safely halt the adjustment calculations and exit in accordance with CLI-REQ-1 before logging the error to the system audit trail.

b. Backward Traceability (rationale for the requirements and the source – RFP and which section in it, client meeting and which notes from that meeting, etc.)

i. Source

- FRP Section 5.0 (Intended Users of the System) - Primary Stakeholders
- RFP Section 7.0 (Known Constraints)
- Client Meeting - 02/09/2026

ii. Rationale

The RFP mandates that the system must serve “Potential Buyers and Researchers” by allowing these actors and users to input information, the

Adjustment Logic Model allows greater degree of personalization to compare property listings in an extremely standardized manner.

c. Forward Traceability

- i. Integration Test to compare a sample baseline FVE, run a test client profile against it and verify that the client-adjusted FVE has accurately accounted for feature weighing.
- ii. Unit Test to see if adjustment calculations are logical and accurate.

Use Cases:

- ADJ-REQ-1: The system shall parse and validate a client adjustment profile CSV, identifying all feature weights and storing them in an active session buffer within no more than ONE (1) second of runtime execution.
- ADJ-REQ-2: The system shall match and augment a baseline FVE for a property based on the active session buffer and return a final "Client-Adjusted FVE" CSV with the delta values within no more than FIVE (5) seconds of runtime execution.

Use Case Flow:

1. User completes a baseline FVE in accordance with FVE-REQ-2.
2. User uploads a Client Adjustment Profile CSV to the system.
3. System triggers data preprocessing in accordance with DSP-REQ-2.
4. System triggers ADJ-REQ-1, consuming adjustment factors into active buffers.
5. System confirms successful ingestion and triggers ADJ-REQ-2, pulling the baseline FVE and performs a secondary FVE.
6. System generates a "Client-Adjusted FVE" CSV report.

User Stories:

User 1

Narrative:

- As a Real Estate Agent,
- I want to adjust the price of a property to better reflect a customer's preferences/needs,
- so that price better reflects the price an individual customer is willing to pay.

Acceptance:

- Given home that needs evaluation and customer personal wants,

- When real estate agent enters weights and home information into the adjustment logic and FVE engine,
- Then an FVE calculation that differs from the default values and reflects the customers' weights is output.

User 2

Narrative:

- As a Potential Home buyer,
- I want to evaluate different homes based on my needs and preferences,
- so that I am shown homes that better fit my wants.

Acceptance:

- Given I have a list of potential homes and am willing to pay extra to ensure that I am getting some attribute,
- When a weight adjusted FVE is calculated,
- Then a FVE reflecting my wants is outputted.

User 3

Narrative:

- As an Operating Manager,
- I want to monitor the variance between the market baseline FVE and the client-adjusted FVE across the entire office
- so that I can ensure that agents are doing their due diligence in leading realistic client expectations and upholding the reputation of the firm.

Acceptance:

- Given that multiple agents have uploaded client CSVs and generated secondary valuations,
- When I compile an aggregate overview for "Client-Adjusted FVE" reports,
- Then the report should clearly display delta values (in dollars) for each client-specific wants.

System Feature 6: Local Data Collection and Storage (DCS)

Priority: MEDIUM

Description: The local data collection and storage of data, i.e. DCS, helps the larger system collect, store, and process local CSV datasets. The DCS makes sure that property data can be

consistently loaded from local directories, validated by internal systems, and preserved for future purposes such as performing audits or making new calculations.

Functional Requirements:

III) DCS-REQ-1: Local CSV Dataset Ingestion

The system must accept residential property data from a CSV file from a user-specified local file path.

- a. Responding to Anticipated Error Conditions
 - i. Missing File: If the selected file does not exist, the system shall stop execution and display the missing file path.
 - ii. Unsupported File: If the input file is not a CSV file, the system shall reject the file and display an invalid file type error.
- b. Backward Traceability
 - i. Internally, the first requirement is intuited based on the assumption that housing data is stored in a CSV format (section 4d) and that Real Estate Agents are uploading files to the system (section 4a).
 - ii. Externally, i.e. from client meeting on 02/09/2026, client client specified that all property data will be provided via locally stored files rather than external APIs.
- c. Forward Traceability
 - i. The requirement can be verified through user acceptance testing when providing valid and invalid file path and confirming correct system responses.

IV) DCS-REQ-2: Generated Report Storage

The system must save each generated report as a CSV file in a configured output directory.

- a. Responding to Anticipated Error Conditions
 - i. File Write Failure: If the system cannot write the output file, it will stop the exporting process and return a file write error.
 - ii. Access Denial: If the system is unable to access the output directory, it will return a storage permission error.
- b. Backward Traceability
 - i. Internally, the second requirement is assumed from the first requirement, given that it is expected that the uploaded file should be stored and processed.
 - ii. Externally, i.e. from client meeting on 02/09/2026, client specified that generated reports must be saved locally in CSV format.
- c. Forward Traceability (how requirements can be verified by users)

Use Cases:

- DCS-REQ-1: The system shall ingest a valid CSV dataset of up to 25 MB in no more than 2 seconds.
- DCS-REQ-2: The system shall save a generated report to the output directory in no more than 3 seconds after calculation is completed.

Use Case Flow:

- a. The user enters a CLI command within the local path to a valid CSV file (DCS-REQ-1).
- b. The system verifies the file exists and is in CSV format (DCS-REQ-1)
- c. The system loads the dataset into memory (DCS-REQ-1).
- d. The system writes the generated report to the output directory (DCS-REQ-2).
- e. The system confirms that the file was successfully written (DCS-REQ-2).

User Stories:**User 1****Narrative:**

- As a Real Estate Agent,
- I want to load a local CSV file of property data,
- so that I can generate an FVE report without manually entering data.

Acceptance:

- Given a valid CSV file within an accessible folder,
- When I run the CLI command with the file path
- Then the system loads the data set and begins processing.

User 2**Narrative:**

- As an Operations Manager,
- I want generated reports to be saved automatically,
- so that I review them at a later time.

Acceptance:

- Given a completed evaluation run,
- When the process finishes,
- Then the system saves the report as a CSV in the output directory.

User 3

Narrative:

- As an I.T. Administrator,
- I want invalid file inputs to be rejected with clear errors,
- so that I can troubleshoot issues quickly.

Acceptance:

- Given an invalid file path
- When the system attempts to load the dataset,
- Then the system stops execution and displays a clear error message.

6. External Interface Requirements

User Interfaces:

This section describes the boundary between the human user (an external entity on your DFD) and the system. It should specify the format, style, and constraints of the interaction.

- CLI
- Ascii menu
- CLI commands (i.e solo FVE <single ppty filename>)

The interface shall be a CLI designed for minimalism, speed, and low resource overhead. The system shall utilize a stdout output (stdout) for successful executions and standard error (stderr) for failure states and error messaging. For ease of use and readability, the CLI will deploy color differentiated outputs, green for stdout, red for stderr, and yellow for non-failing warnings. Users will interact with the system using the standardized POSIX command syntax (i.e `calc-solo --file <filename.csv>`). To minimize the gulf of execution, the CLI will feature a ASCII menu to detail the available commands and program options for quick browsing.

Hardware Interfaces and Requirements (add-ons):

- at minimum run the said docker image.
- Product built for 7th gen Intel Nucs though can easily run again on something with Docker CLI, internet access, and suggested 6gb of RAM, and 256gb memory

If for some reason a Docker image does not work, a separate contract addition may be made to include an ansible image with the sole purpose of declaring any system dependencies and project setup.

Software requirements

Given the application runs in a container with all the dependencies built in, it becomes isolated from the rest of the system. The only setup which will be outlined in a SETUP.md included with the project repo will be to pull the image and run it inside of docker. The image isolation from the rest of the system allows for ease of updates and rollbacks in the event of any later needs. Given these requirements, the OS it runs on doesn't matter (again as long as it can run Docker images).

The next few sections outline baseline minimum requirements for the agreed upon product. However, at the end of each is a said add-on that may be renegotiated as an add-on per said contract if desired. These may be additional tasks that can be added at a later date.

Internal data: (add-ons)

Cached along with the docker image. Given we don't expect internal data to be larger than 25 mb, we have defaulted to using CSV files to load in datasets. However, if desired, a mongoDB container used for storing said dataset files may be created. This would again be another add-on to the current contract that would need to be discussed.

External Connections & Networking (add-ons)

If desired outside of product scope, external outside network connection to query the given CLI app may be provided. This will require a separate add-on contract and should be discussed if desired as an add-on. This would include

- Connecting existing application to client's meshnet/tailnet
 - o In the event one is not setup, we are able to set one up. However, this would be under an additional contract service and should be discussed separately.
 - o Advising on tailnet policies to use said application with given port asecss
 - o (Creation of meshnet/tailnet for client is under separate contract)
- Configuring Docker image such that only one port is exposed via the tailnet traffic
- Uptime tracker via beszel or Portainer

7. Software Quality Attributes

Performance Requirements

All requirements below are assumed to be on a generic intel nuc NUC12WSi5 as outlined by [this](#) specification document.

Performance Requirements-1: Single property

The FVE engine shall calculate and generate a single-property estimate report in CSV format in no more than 5 seconds from the moment the user submits the parameters.

Performance Requirements-2: Multiple properties

The FVE engine shall allow a batch of no more than 100 properties to be calculated as a batch job for neighborhood-level analysis in no more than 30 seconds.

Performance Requirements-3: Max file size

The application shall support data processing of individual CSV files no more than 25 Megabytes (MB) in size.

Performance Requirement-4: Processing power

The FVE shall not consume more than 100 MB of ram at any given time unless unrestricted mode is on (no RAM consumption limit).

Reliability Requirements

Reliability Requirement-1: Stability and congruence under stress

Ensure that when running a benchmark stress test, container does not crash under large request amount of requests. Files parsed can be up to 25MB and will throw a large file exception if exceeds the 25MB. If the overall system RAM usage cannot handle the current container, the container shall send a notice to the user requesting to shut down notice to another program using too much memory for the container to run effectively. If the user does not take any action the container shall shutdown within 2 minutes.

Reliability Requirement-2: Maintaining batch processing for data integrity

Each batch FVE generation must record any violations amongst one or more properties in the CSV file, resulting with the system skipping the process and log the batch failure and reason to continue onward other valid records without interrupting the cycle flow. It will then tell the user what each issue is and where to fix them inside said file. The JSON config file may be changed to throw an error instead of skipping the process. This shall be in the settings command.

Maintainability Requirements

Maintainability Requirements-1: Modularity

The Adjustment Model Factors must be decoupled from the FVE core calculation engine, existing in a separate CSV/config.json file to allow feature weights to be updated without recompilation of the system source code.

Maintainability Requirements-2: Coding Conventions and Documentation

All Python modules shall follow current contracted company standards and include docstrings to ensure ease of onboarding for new users. If no standards are in place, Python Black shall be used as a code styler, and Flake8 as a linter. Basic documentation shall be held on an internal pydoc page which is generated from internal pydoc comments.

Maintainability Requirements-3: External Configuration

External configuration shall be handled via a JSON config file. This file is to be updated via the CLI via a `/settings` command. User do not need to directly interface with the JSON file and shall instead simply edit their settings via the `/settings` commands. Settings may be used to change change factor weights, error weights (verbose, light, etc), data storage (CSV/TSV/etc) and other previous items listed in the aforementioned sections on settings. Settings may be imported and saved between images for ease of exchanging as an option inside the settings page.

Maintainability Requirements-4: Descriptive Error Messages

The system shall display detailed error messages containing the type of error, the failed step, and the affected module, along with a brief possible fix. Each error message shall be documented in a JSON object. Error types shall be explained in pydocs.

Maintainability Requirements-5: Feature Rollbacks

The system shall support rolling back to previous working versions of the application. This shall be done via docker image version tags. User may use per image version tag for quick rollbacks to any desired version. To update users will simply run docker compose for the latest version for a new image.

Availability Requirements

Availability Requirement-1: Uptime

The locally hosted CLI application, if run on minimal hardware specification(s) of 12th Gen Intel NUC 12, must be available on a realistic scalability of 95% within operable business hours provided by the dependency of the user's workstation operational runtime. Uptime shall be moderately dependent on system specifications and requirements while nothing else is running from the background.

Availability Requirement-2: Startup time initialized

CLI application shall be fully initialized and ready to begin accepting user inputs within 5 seconds after being launched on an issued workstation for meeting the minimum hardware specifications. This does not include the container-image creation time from docker and assumes that the container has already been made.

Container creation time may vary but shall not exceed 5m. This assumes stable internet connection with at least the following:

- 20.8 Mbps download
- 6.20 Mbps upload

Slower internet speeds than this will work but is not guaranteed to exceed 5m.

Scalability Requirement

Scalability Requirement-1: Capacity processing for localized project integration.

The FVE engine utilized for the project/integration is for local, single workstation to run up/execution, and thus, not required to support scaling or concurrent multi-user access.

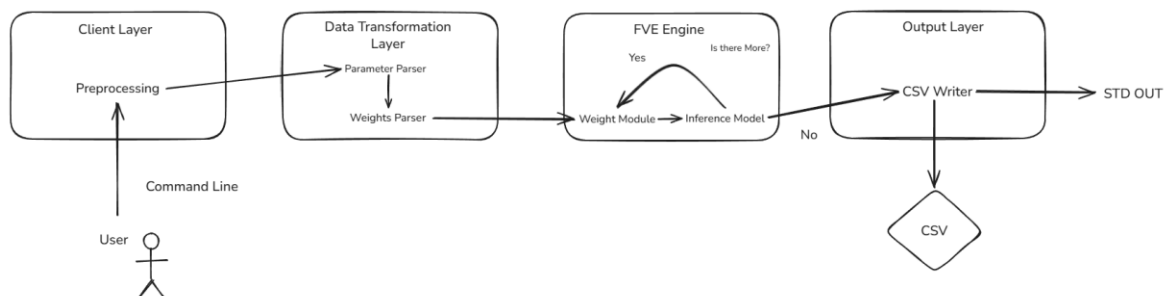
Scalability Requirement-2: Batch size must be within limit ceiling

The FVE engine can support a maximum batch size of 100 properties. Any input exceeding this limit must be rejected and clearly display an error message informing the user that the maximum input has been exceeded. The ceiling ensures that the local hosting setup has been addressed and can be further upgraded upon infrastructure systems for cloud migration if needed.

If a higher ceiling is required, or connection to the same data storage files are required, a separate contract shall be formed. This would include a Mongo DB container and networking setup as per outlined in Section 6 add-ons.

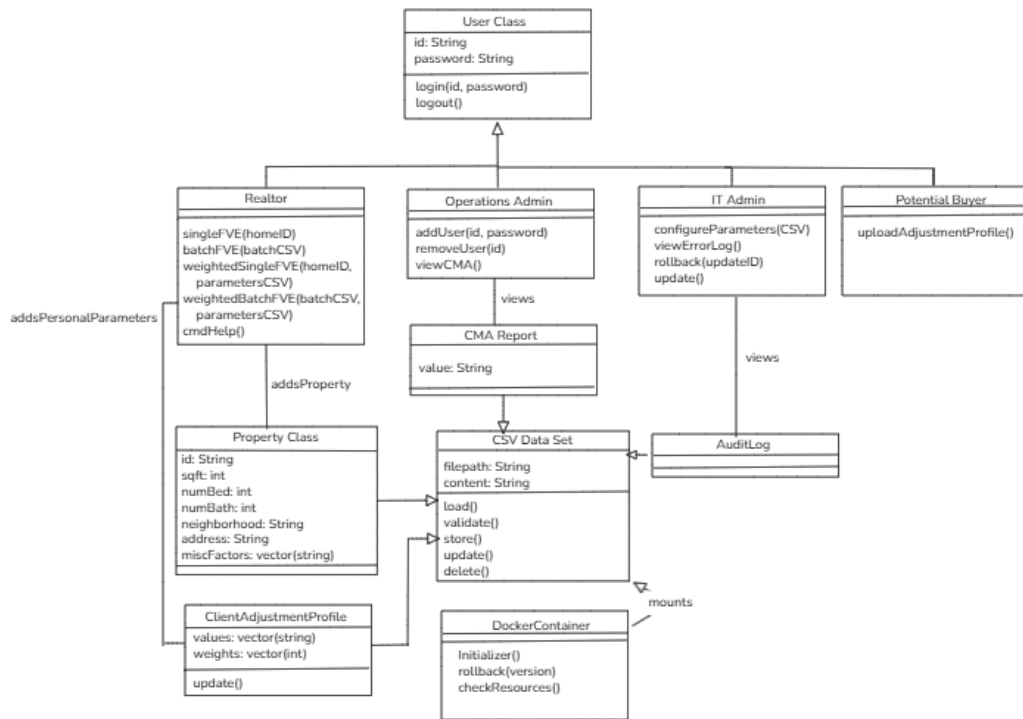
8. Analysis Models

Architecture Diagram



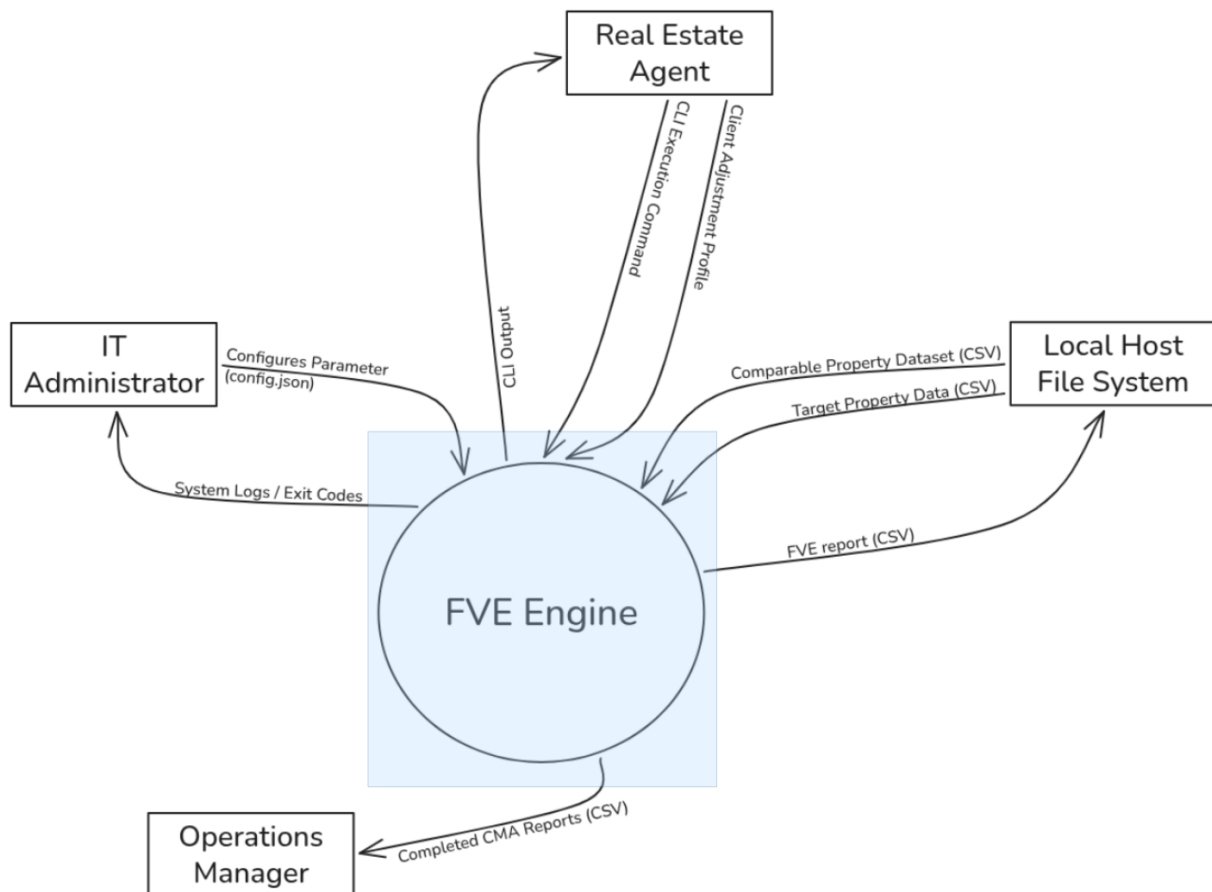
In this figure, the different components of the system are shown and how they interact with each other. The user will insert their respective inputs (through the command line) in the client layer. Then their data will be processed into tokens in the client and then parsed in the data transformation layer before going into the FVE Engine. The FVE Engine is the main determinant where the CMA estimated value will be generated, depending on the various criteria of rulings to achieve a better-defined value. The CMA will be constantly looped for any properties until the batch is complete. Finally, in the output layer, the CMA will be imported as its output and then applied onto the CSV file for the result.

UML Class Diagram



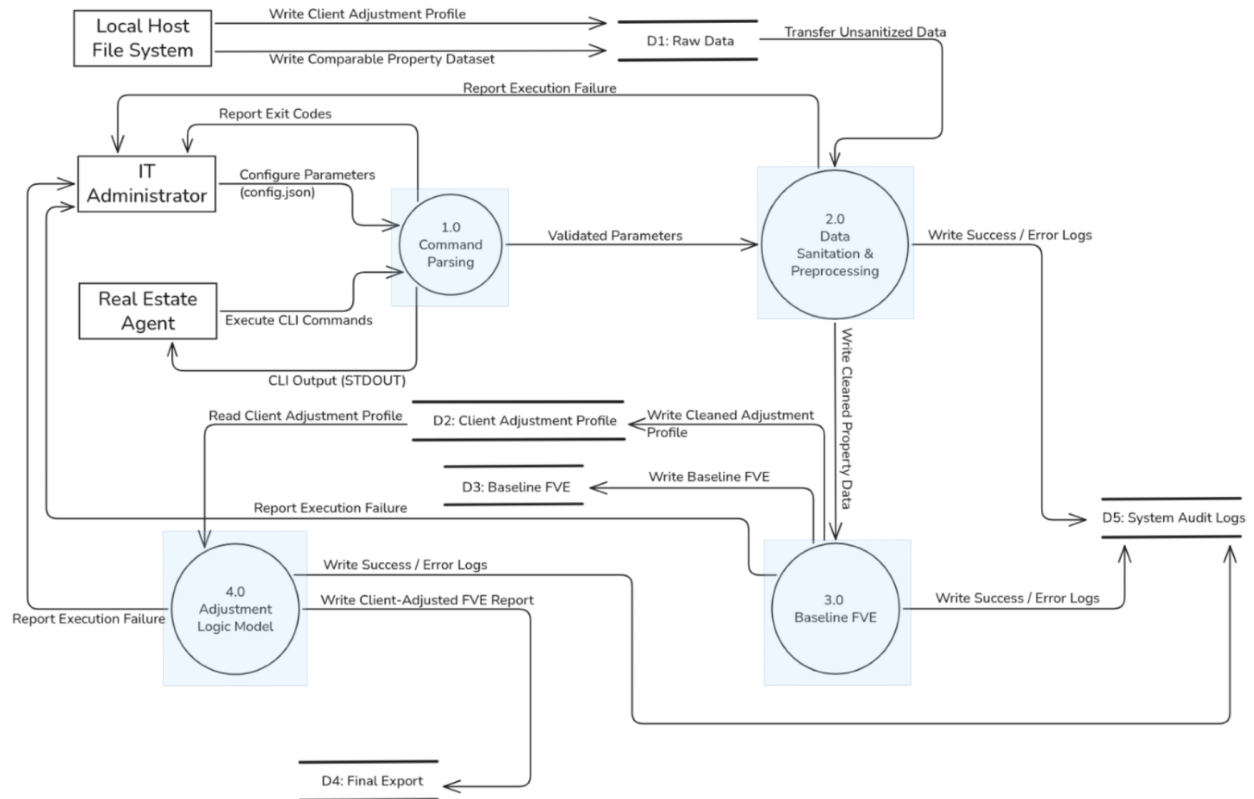
Shown are the various user classes and CSV file types, and how they interact. There are 4 user classes : realtor, operations admin, IT admin, and potential buyer. There are 5 CSV files: the property class, CMA report, audit log, docker container, and client adjustment profile.

Context-level Diagram



The figures show all of the connections to the FVE Engine. This is to show how data flows into and out of the FVE Engine. The real estate agent, the primary users, inputs the adjustment profile of the individual customer and inputs the execution command. In return, the FVE outputs the CLI output of the execution. The local host system provides the dataset of the actual property and comparable datasets to the FVE Engine. It is given the FVE report in CSV format to be stored in logs. The IT administrator configures the parameters and receives the exit codes. The operations manager receives completed CMA reports.

Data Flow Diagram (1 of 2): Running FVE from start to finish (including Client Adjustment)



D1-D4 represents sub directories within the database and has been explicitly stated for added clarity for the data flow diagram.

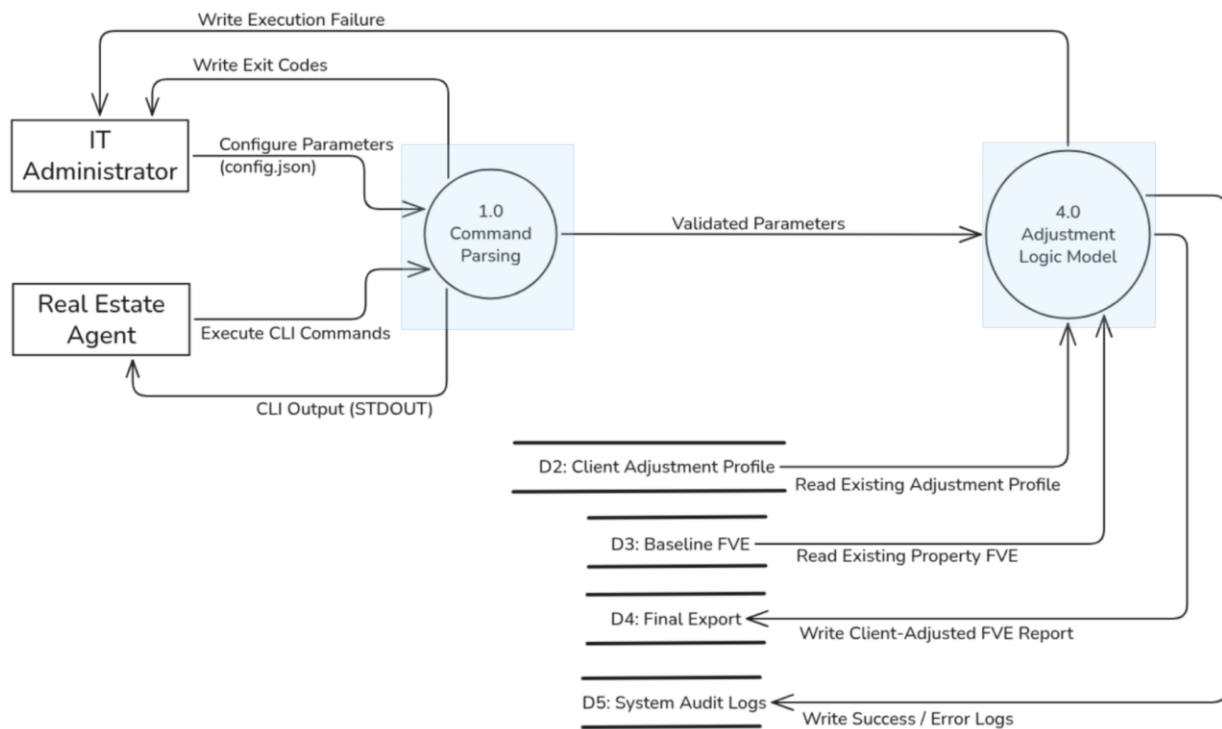
For the figure, the FVE Engine is clearer with better data flow interaction amongst how the actors of the Local Host, IT Administrator, and the Real Estate Agent are interacting. The Local Host is inputting the original file for raw data to be processed. The IT Administrator ensures configuration parameters through config.json as well as receiving any execution code failure reports from the transmitted processing operation. The Real Estate Agent may execute CLI commands to parse and retrieve the output through STDOUT.

For each central processes, it helps perform the following:

- *1.0 deals with command parsing through CLI commands from the Real Estate Agent and configuring parameters from the IT Administrator. These inputs are taken and are validated to be processed to 2.0 .*
- *2.0 will take in the validated parameters and the original, raw data to return its property data to 3.0 and transfers its clean profile to D2. Any other log successes or failures throughout the process will be navigated to D5.*
- *3.0 is the baseline for FVE where it takes 2.0 cleaned property and operates the calculations. The output will be passed to 4.0 and other FVE write baselines to D3. Any other errors or failures logged will be passed to D5.*

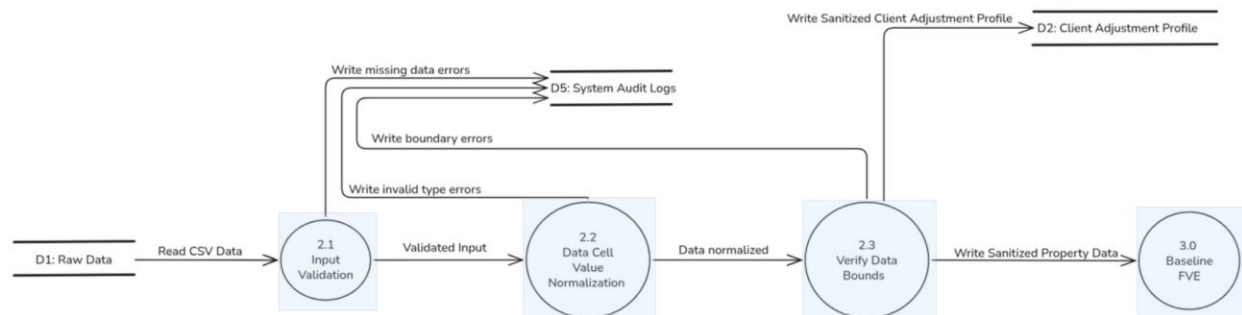
- 4.0 operation would take in D2's profile returns and applies its client-specific adjustment logically models to its finalized FVE report. This report will be passed to D4, any written model logged failures or successes will go to D5, and the execution of failures be passed to the IT Administrator.

Level-1 Data Flow Diagram (2 of 2): Running Client Adjustment on preexisting properties with prior FVE



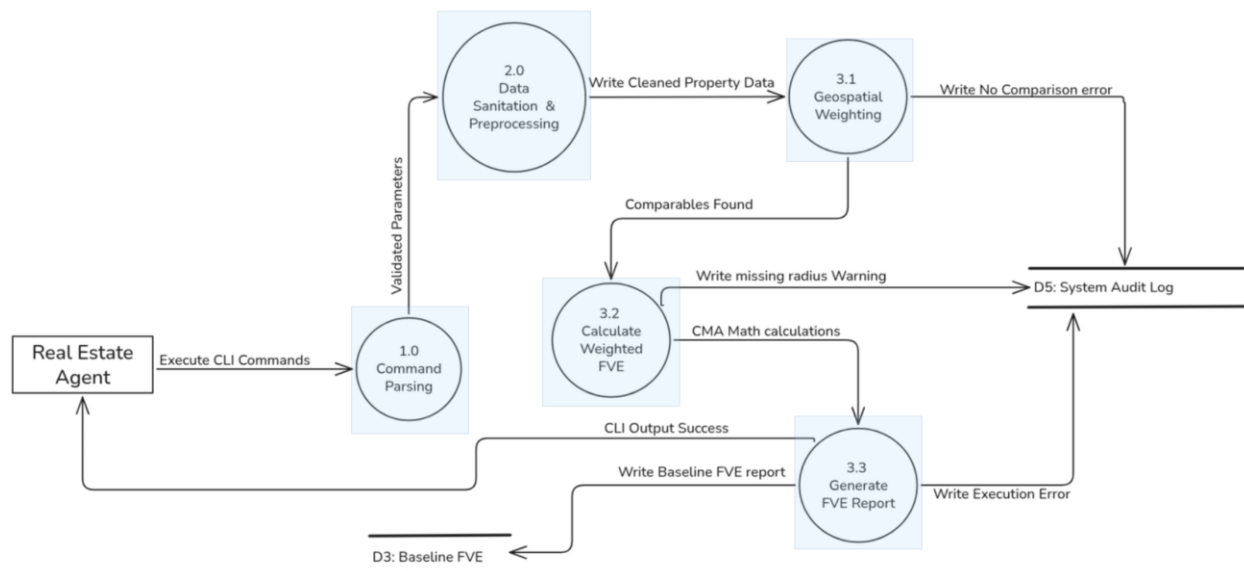
This figure shows a different approach where pre-existing data properties are already present for the FVE Engine to operate. The IT Administrator and Real Estate will pass their respective configuration parameters and CLI Commands to 1.0 as before. However, the validated parameters will now be sent directly to 4.0 ALM. With the ALM's given pre-existing data, the logic will read existing profiles already given from D2 and D3 FVE baseline properties. The ALM will perform the same operation and write its finalized report to D4 and logging it to D5. Any execution of failures or errors will be written to the IT Administrator.

Level-2 Data Flow Diagram (1 of 3): Data Sanitation & Preprocessing Process



The figure dives into the Baseline FVE Process and how its sub-directories processes are sanitized and processed from 2.0. The raw data gets passed into 2.1 - 2.3, where each operation is checked for any boundary errors, invalid input criteria's, and missing data that's necessary to proceed. Each operation will be first validated, normalized, and then verified to ensure that 2.0's core functionality will be sanitized and standardized to be passed to 3.0. Any raw data that violates the errors will be directly sent to D5. For 2.3, its outputs will be judged and redirected to D2 for written sanitized client adjustment profile or its sanitized property data be passed to 3.0.

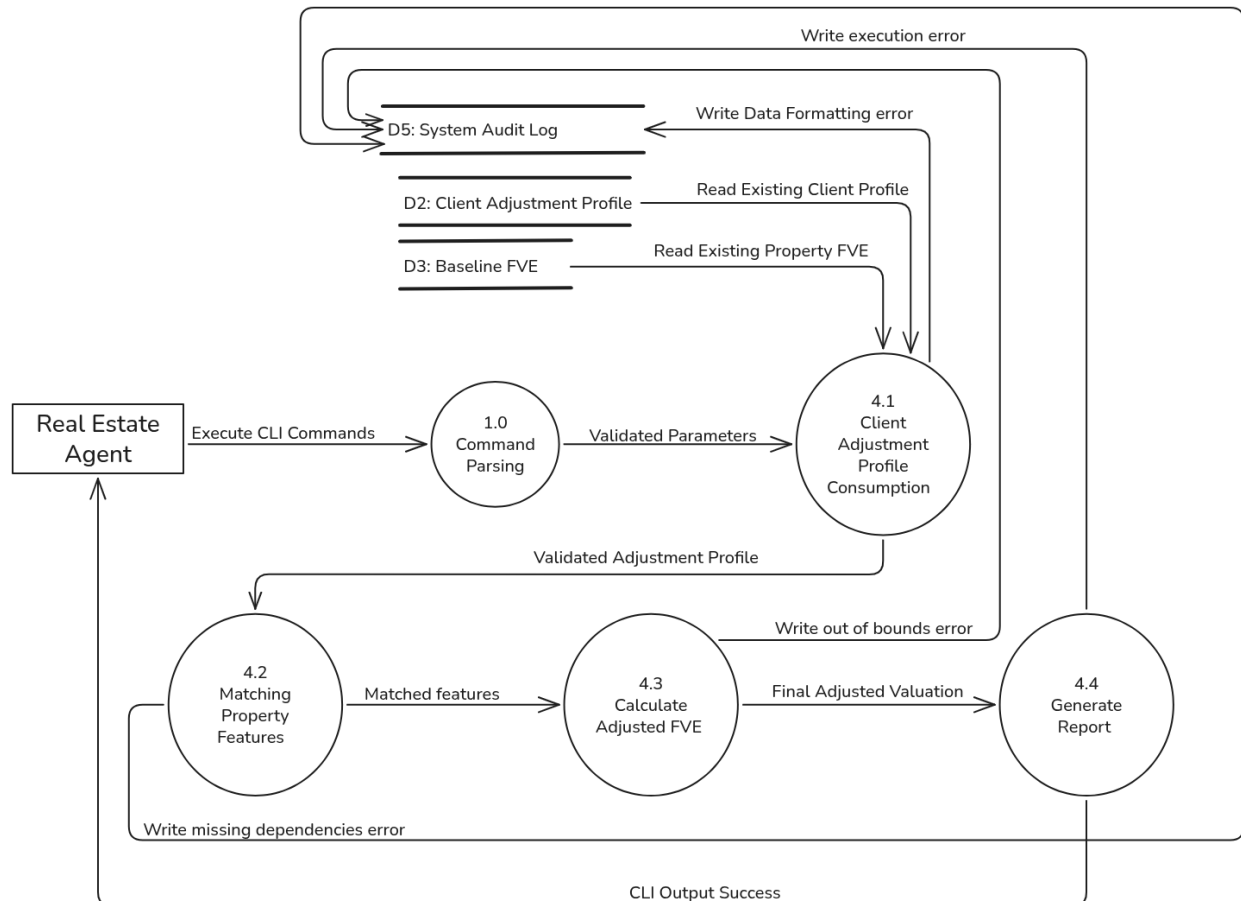
Level-2 Data Flow Diagram (2 of 3): Baseline FVE Process



The figure deals with 3.0 for its Baseline FVE Process. After 2.0, its cleansed property data will be redirected to 3.0. The geospatial weighting will be first encountered, where it weighs various comparable properties that the geographic criteria are applied onto. Such comparable properties found will be passed to 3.2 while any missing will be sent to D5. 3.2 will read the parsing comparable and weighted data and run the CMA calculations to get an output. If the radius goes missing for any comparable calculations, write it to D5. Otherwise, the calculations will be passed to 3.3. The final step for 3.3 will take in the CMA calculations and produce the

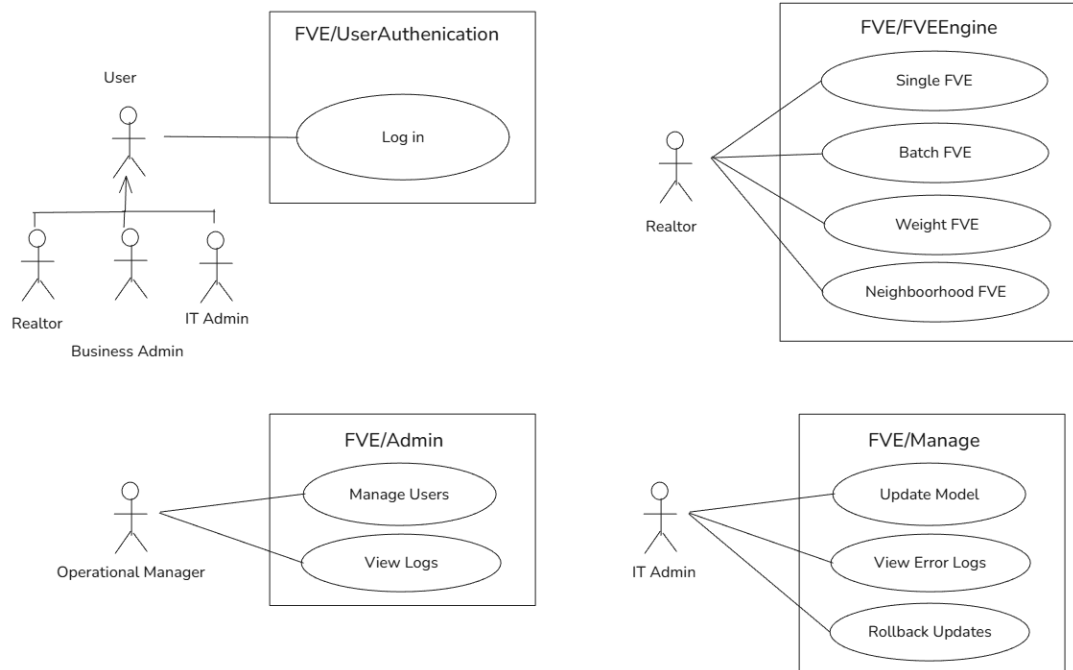
FVE. Upon success, it will write the FVE report to D3 or D5 if the execution was unsuccessful.

Level-2 Data Flow Diagram (3 of 3): Adjustment Logic Model Process



The figure dives deeper into 4.0, where the diagram shows 4 different sub-sections for processing. For 4.1, it retrieves all 3 sources from the Real Estate Agent, D2, and D3 to validate and merge all information into a CAP. This profile will be passed to 4.2 while any formatting pathway errors are dragged to D5. Sub-section 4.2 takes the whole CAP and matches each user's subject property features, which upon succession will be passed to 4.3. Any missing dependencies found from matching calculations will be audited on D5. For sub-section 4.3, the matched features will be applied to the adjustment logic to compute the FAV and be transferred to 4.4 to generates it outputted report and send it to the Real Estate Agent. Any boundary or execution issues arising from these sub-sections will be reported back to D5.

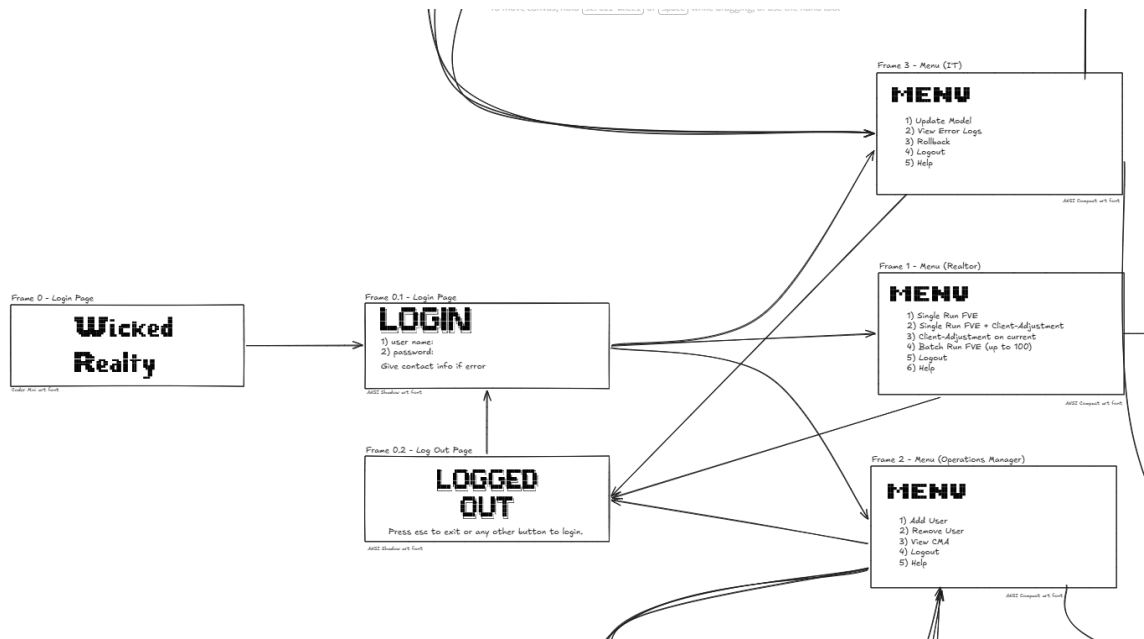
Use Case Diagram



The figures dive into the primary actors that are interacting with the Fair Value Estimate (FVE) and the specific actions they are authorized to perform within those actions. First the Fve/user authenticator creates a base level of access to the system through a log in function. This base layered pathway is then inherited by the systems three sub-actors the realtor, business admin and the IT admin. The diagram then branches into role specific capabilities, showing the realtor interacting directly with the FVE. Within this module, the realtor is allowed to execute core processes specifically a single FVE, a Batch FVE, weight FVE and a neighborhood FVE. The Operational Manager accesses the FVE/Admin module. This section grants the manager administrative privileges to Manage Users and View Logs. Finally, the IT Admin interacts with the FVE/Manage module to handle system maintenance. Through this module, the IT Admin is authorized to perform technical support tasks, including the ability to Update Model, View Error Logs, and Rollback Updates.

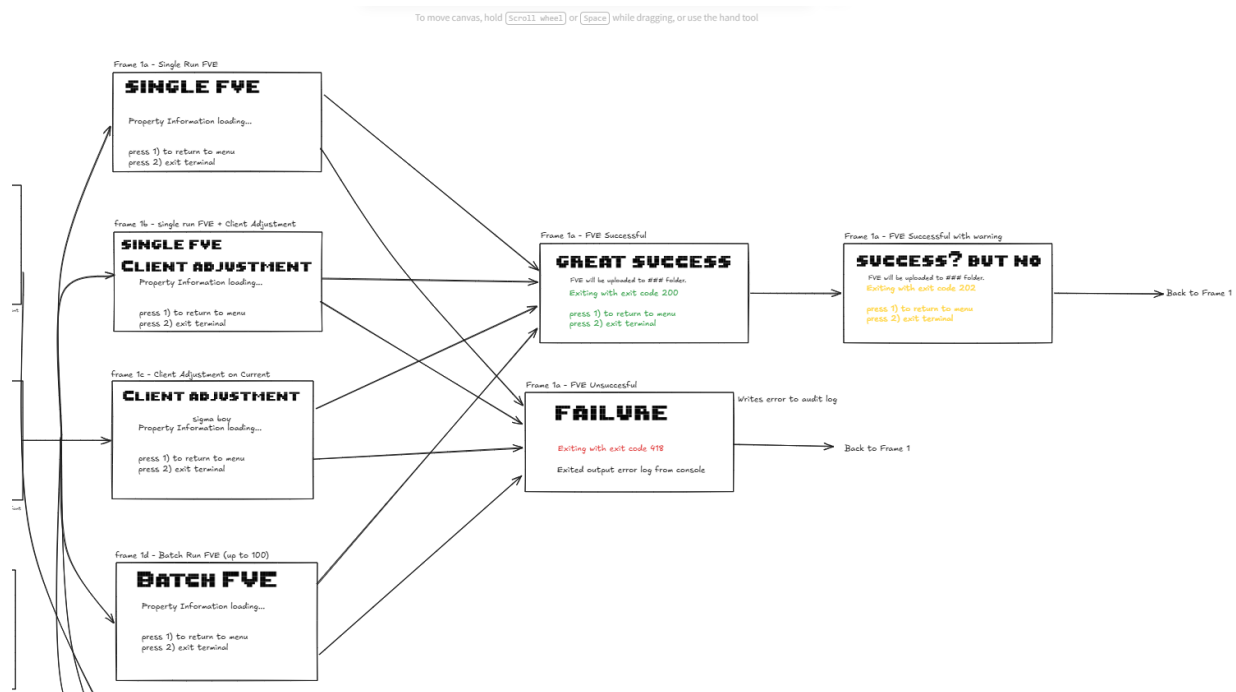
9. Detailed System Design

Log in screens and menus:



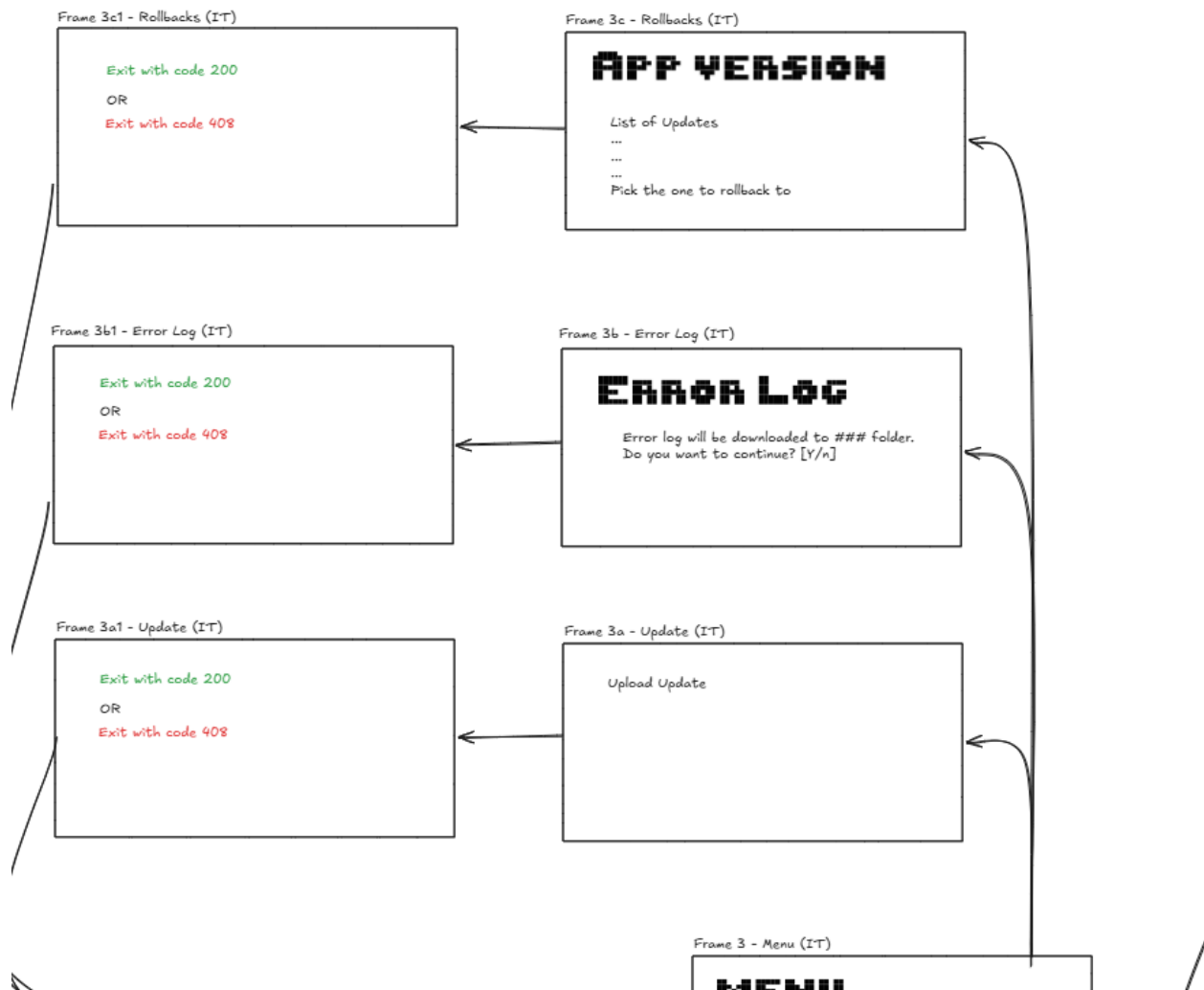
This is a prototype of the login screens shown to the client. It features a title screen, a login screen, a menu screen for each role, and a log-out screen. These would all be in the command line, as mentioned in System Feature 2: CLI application, for ease of use, as would every other screen shown in the prototype.

Realtor Screens:



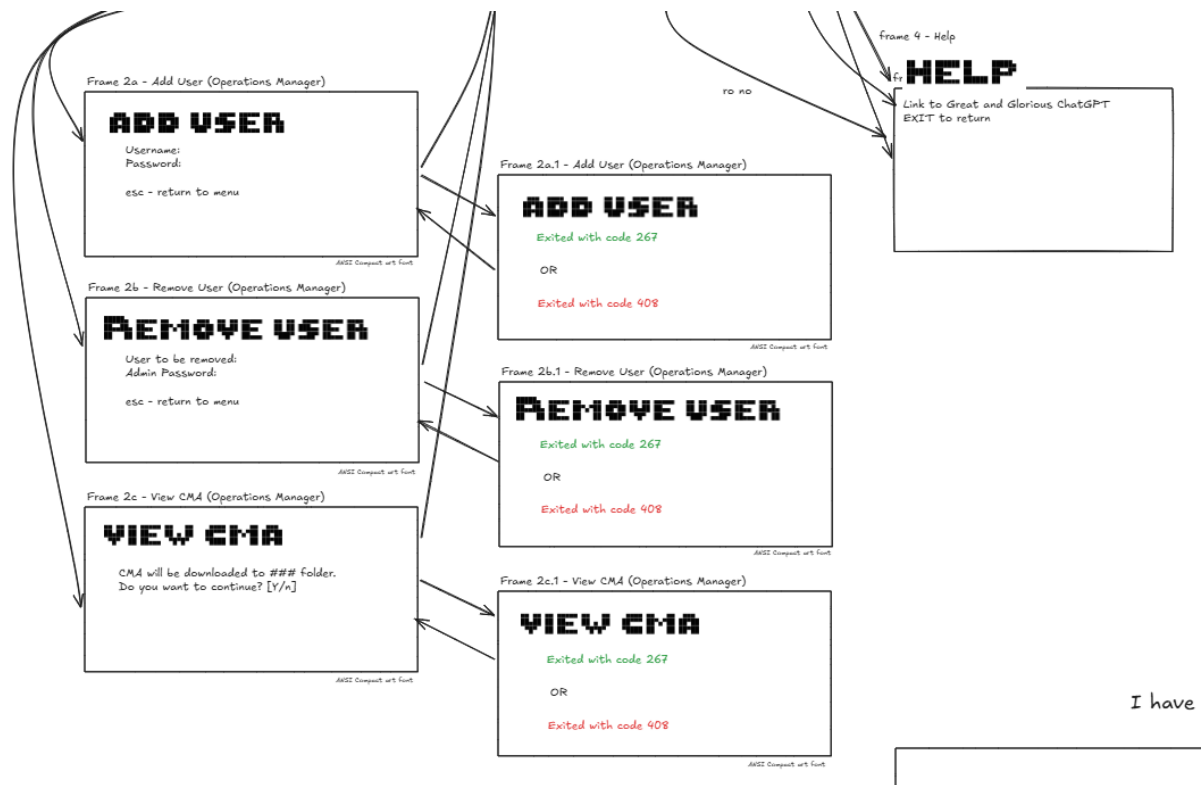
Here are the different options for the realtors on the same prototype. The realtor would be able to run a single FVE, a single FVE adjusted for the client, or a batch FVE. To do this, the user would upload a CSV into the `INPUT_FILES_HERE` folder, and if adjustments were necessary, they would upload a CSV to the `ADJUSTMENTS` folder, as per the requirements of System Feature 6, Local Data Collection and Storage, and System Feature 5, Adjustment Logic Model. In the background, the data would then be processed and parsed as per System Feature 3, Data Sanitation and Preprocessing. Afterwards the FVE engine would calculate the FVE, as said in System Feature 4, FVE Engine. This would store the CSV in an output folder accessible to the realtor, as noted in System Feature 6, Local Data Collection and Storage. These would all lead to a success screen or a failure screen. As mentioned before, this would all be done in the CLI as per System Feature 2, CLI application.

IT Screens:



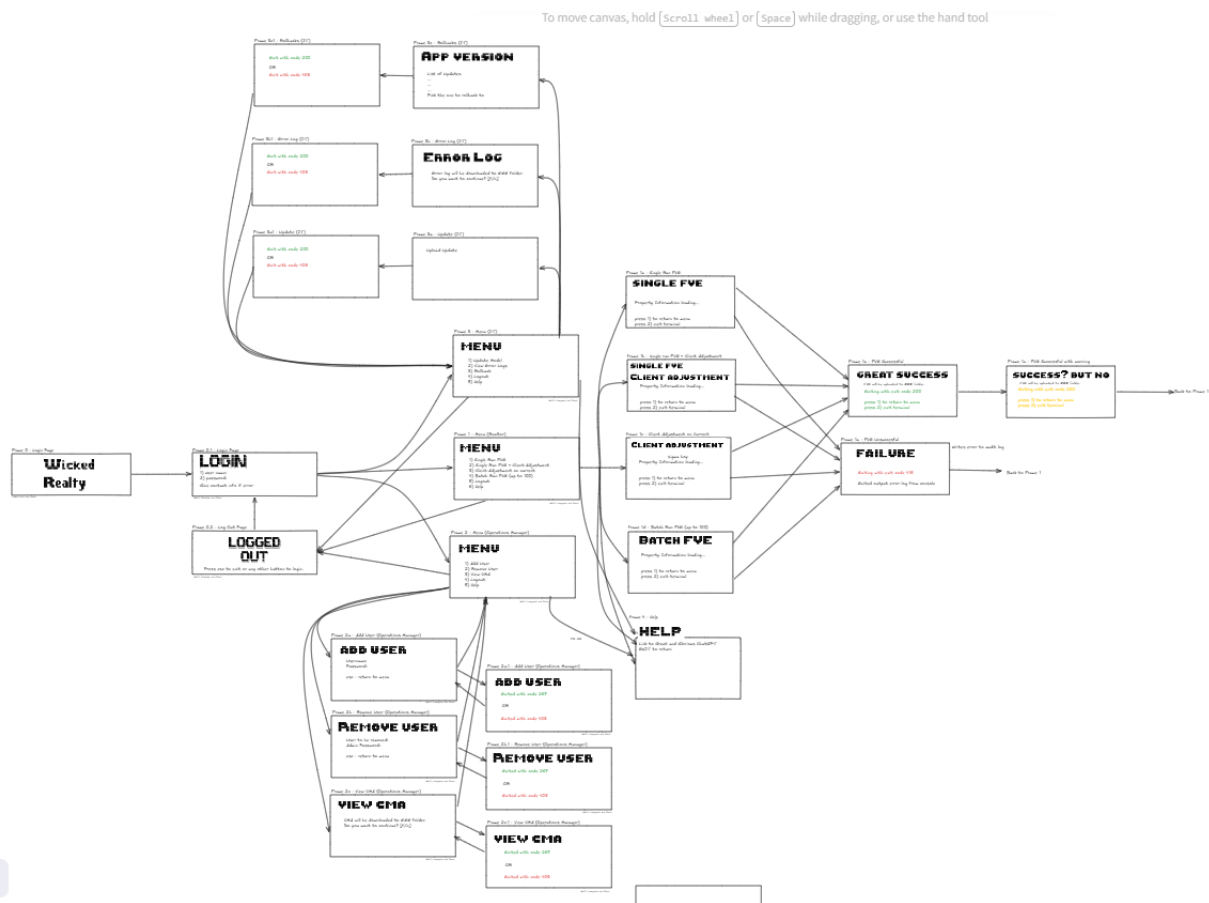
The IT admin screens are the app version, error log, and an upload update screen in the prototype. The error log was the most important, as any failures would be recorded there and output to an error log CSV, as per System Feature 6, Local Data Collection and Storage.

Operations Manager Screens:



Operations Manager screens in the prototype show Add and Remove Users and View the CMA. The CMA would be viewed locally on a CSV as per System Feature 6, Local Data Collection and Storage.

Overall Diagram:



Here is a fully zoomed-out diagram of the UI prototype and all of its screens to give a better picture of the full application and how any user could navigate the app. All of the screens would be shown in the CLI, as per System Feature 2, CLI application.

Hardware, Software, and Communication Interfaces

To create this CLI interface, Ink's library will be used. This allows for the screens with colors and fonts to indicate visually successes, failures, and errors. In addition, we xterm-256color was chosen as the emulator of choice within each docker container to render the ink colors.

Hardware Interface: T

ested on NUC11PAHi7 with 16 gb of RAM, though any device that can run docker should suffice.

Disk Space: 100mb not including docker.

Software Interface:

A docker compose file is provided alongside a Docker file to build and update the container. Update deployments can be run within the container via a simple git pull and revert with git. The docker compose file houses several volumes; of note are the following: FVE_OUTPUT, DATASET, INPUT FILES, TMP, ERROR_LOGS, config.json, adjustments.json, users.json, roles.json/. This allows the container to retain any important files as persistent memory in the event of a container crashing or container being brought down.

OS:

Any OS compatible with downloading Docker and running the container. For development, prefer Ubuntu 24.04. If the client wishes, upon additional fee an ansible image may be setup per each operating system for ease of modification, though currently is not provided due to the user side of the application being done purely through docker.

Remote access:

To test remote access, we used an internal tailnet with the same ports as defined in the docker file. We then connected via another pc to the said device running the container at the given port in the docker file. The UI was made with ttyd and ported to the docker port which appeared on devices web url. However, as previously stated, a full tailnet can be provided upon an additional contract fee and discussion and is assumed to be already within the client's system.

Runtime Environment:

Node.js 16 inside of Docker

Dependencies: as listed in my-cli/package*.json

"csv-parse": "^6.2.1",

"csv-stringify": "^6.7.0",

"ink": "^5.2.1",

"ink-select-input": "^6.2.0",

"ink-spinner": "^5.0.0",

"ink-text-input": "^6.0.0",

"meow": "^11.0.0",

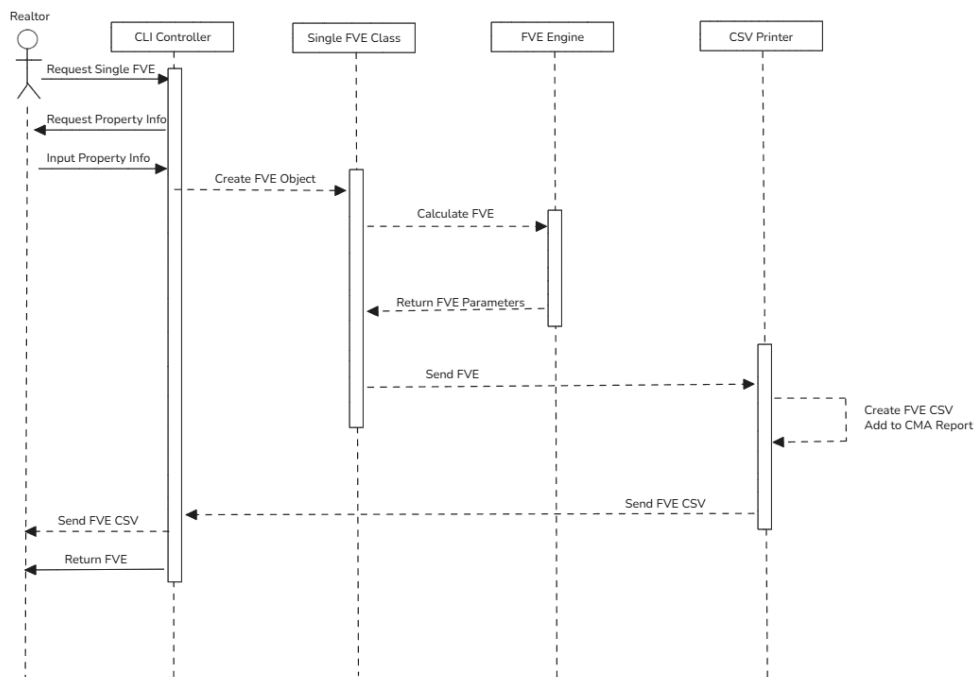
"react": "^18.2.0"

10. RD 3.0: More Diagrams

To preface this section, most of the diagrams were created before or during the creation of the FVE engine. Because of this, the sequence diagrams and much of the other diagrams were based on the prototype rather than the final application. This led to some discrepancies in classes and overall data flow between the diagrams and the application. Anything that does not match the application should match the prototype.

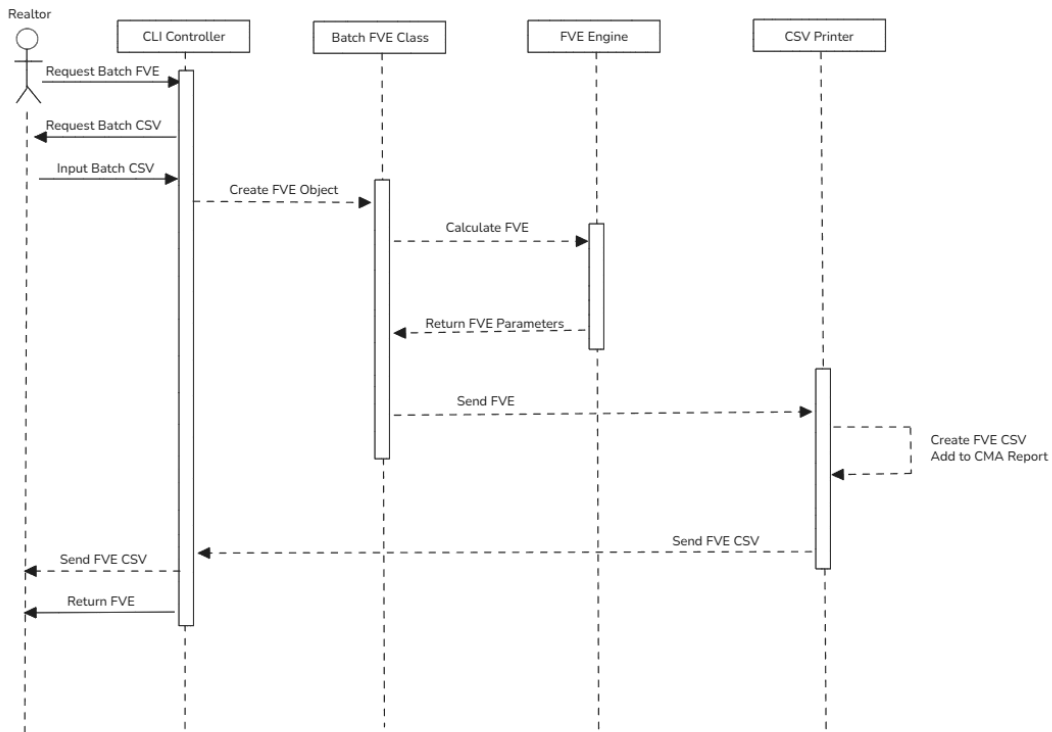
10.1 Sequence Diagrams

Sequence Diagram 1: Realtor Single FVE



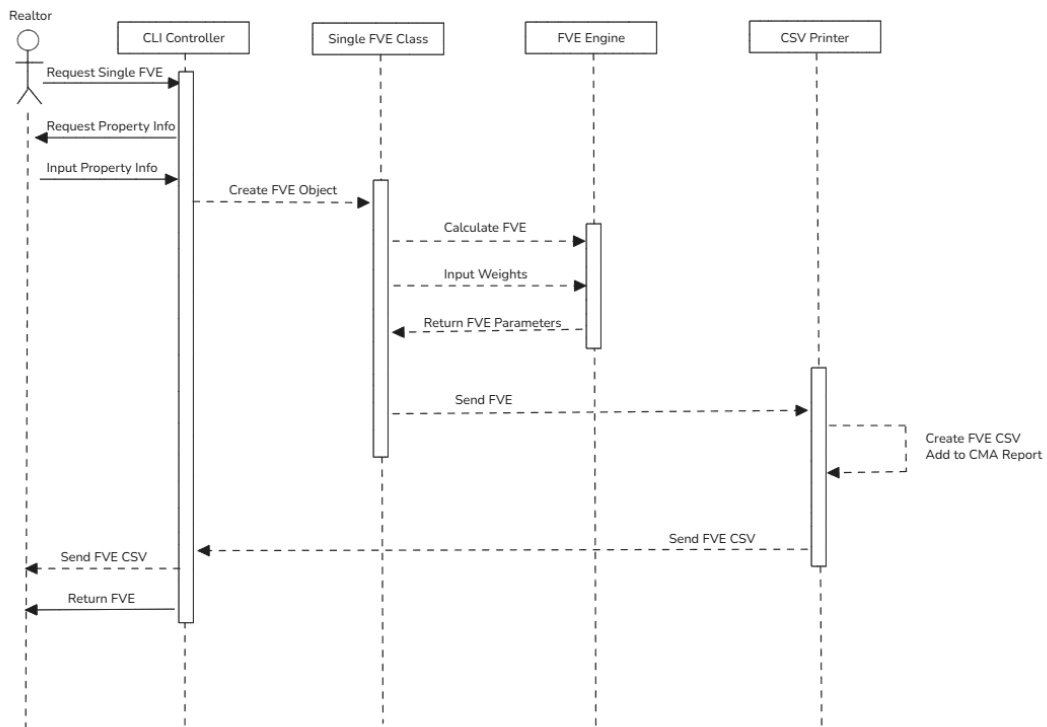
Here is a sequence diagram of the realtor single FVE. This shows how the classes interact with each other to calculate the single FVE. First, the realtor inputs the CSV, which is brought to the FVE engine, where the FVE is calculated. This is added to the CMA report, then turned into a CSV. This is put into an output folder for the realtors to access.

Sequence Diagram 2: Realtor Batch FVE



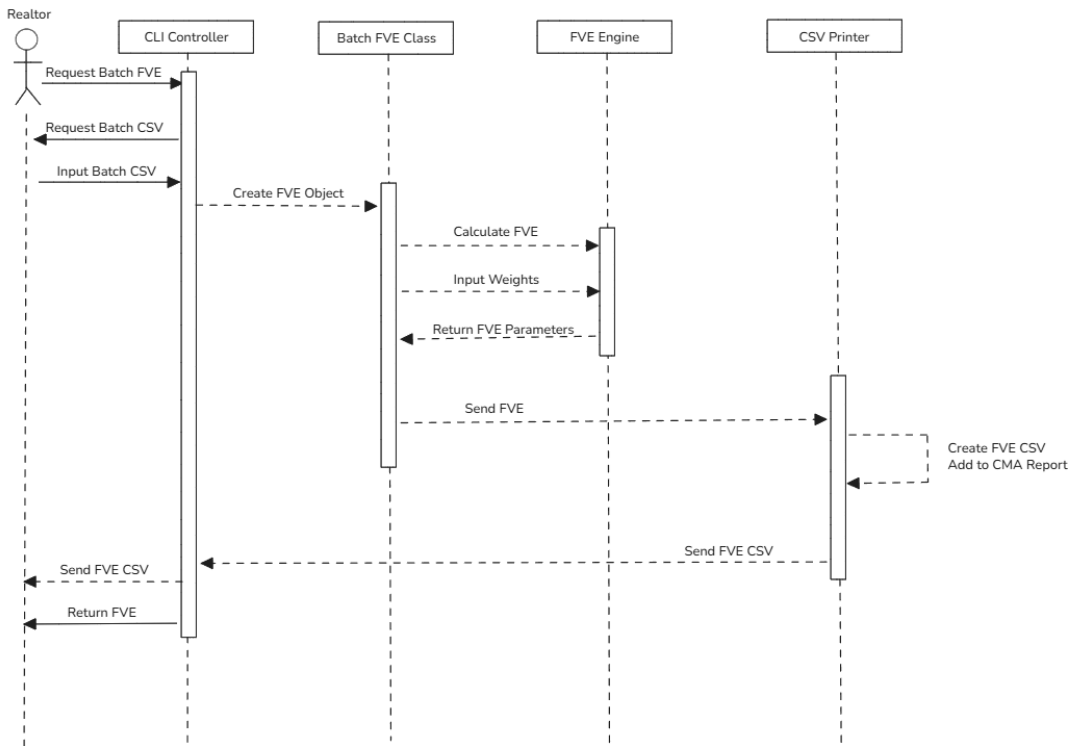
This is the sequence diagram of a batch FVE. It works mostly the same as the single FVE. The realtors input a CSV of the properties, which the engine then calculates all the FVEs. This is written to the CMA, then written to a CSV in an output folder that the realtors can use.

Sequence Diagram 3: Realtor Single Weighted FVE



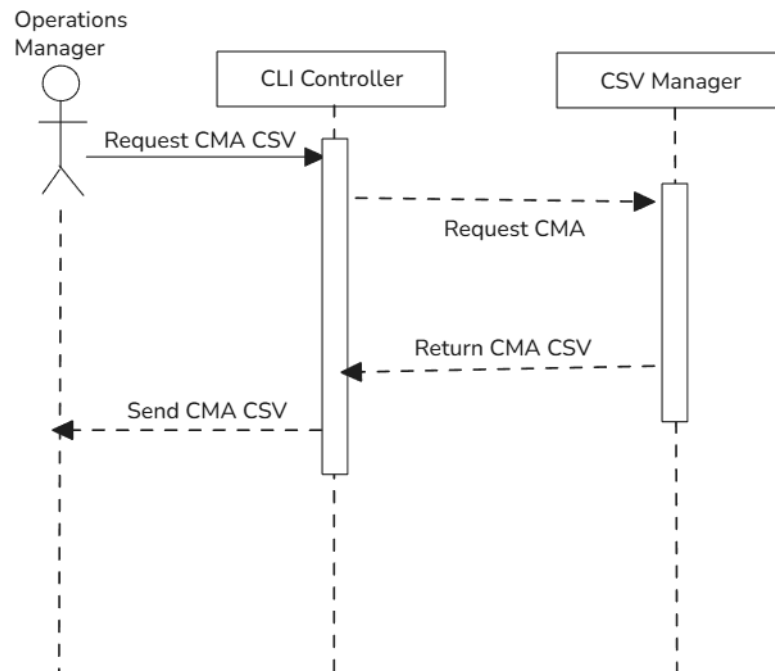
Here is a single weighted FVE. The only difference from the unweighted single FVE is that the FVE class now sends the input weights to the engine. Everything else remains the same.

Sequence Diagram 4: Realtor Batch Weighted FVE



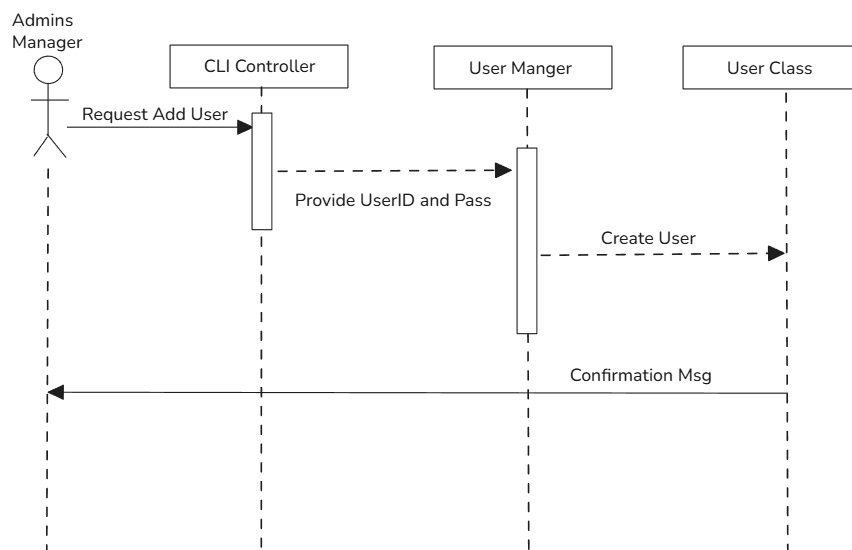
Here is the batch weighted FVE. It works the same as the unweighted batch FVE, with the exception that the batch FVE class now sends the client's weights.

Sequence Diagram 5: Operations Manager Views CMA



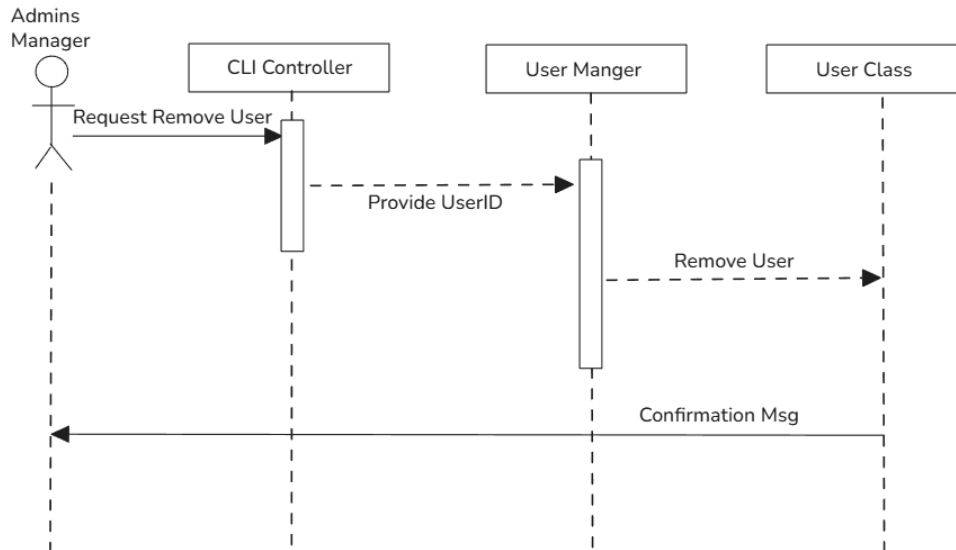
Here is the operations manger viewing the CMA sequence diagram. The operations manager clicks on the view CMA button, which sends a request to the CSV manager. This then returns the directory where the CMA is located to the user.

Sequence Diagram 6: Add User



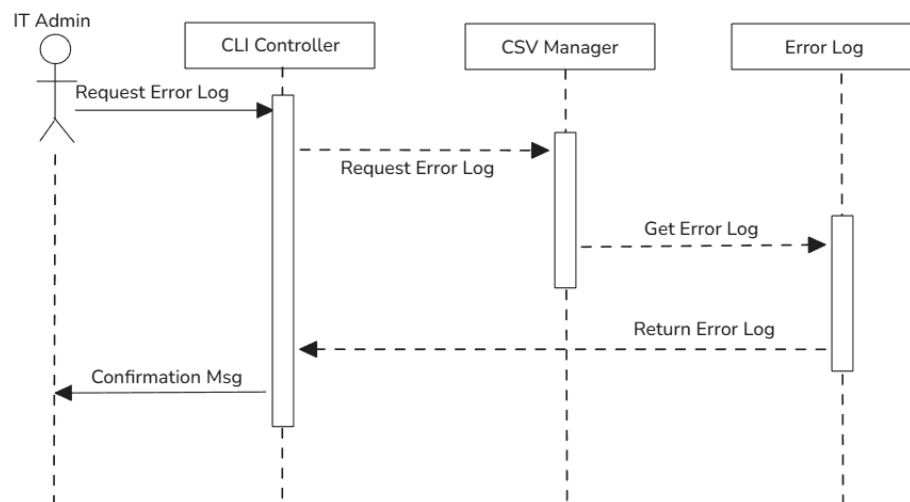
Here is a sequence diagram of the manager adding a new user. The manager provides the new username and password, which is then sent to the user manager. This then updates the user class, and a confirmation message is sent to the manager.

Sequence Diagram 7: Remove User



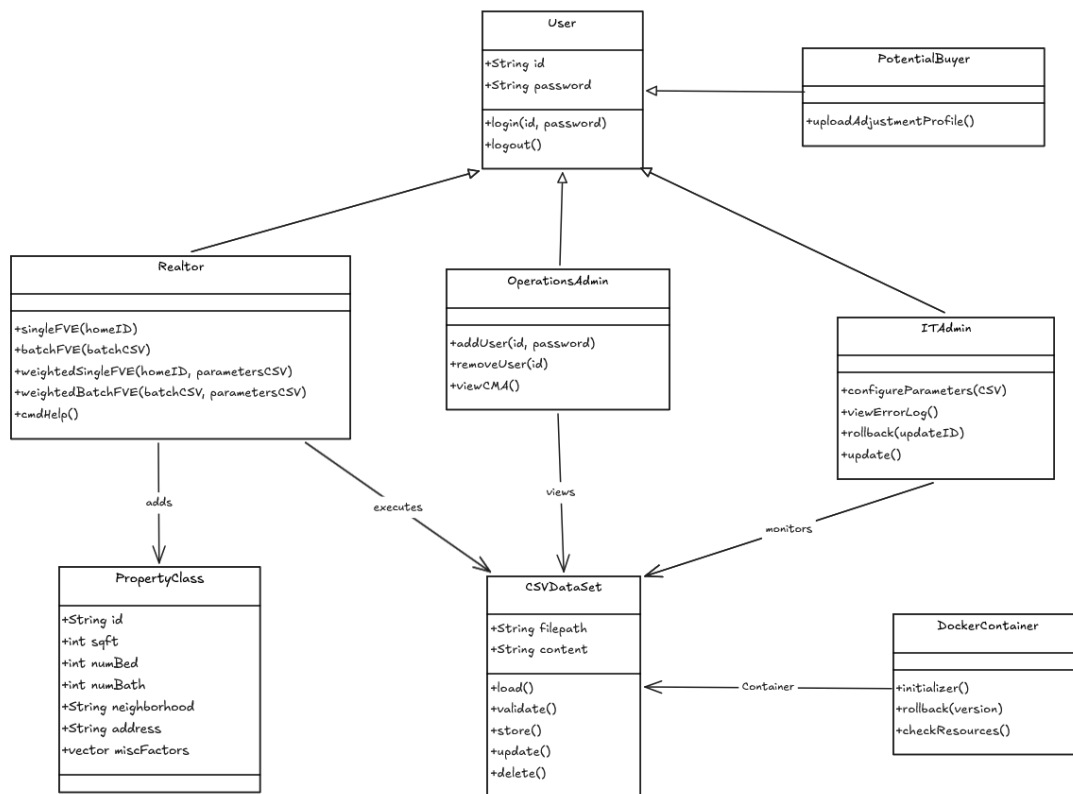
Here is a sequence diagram of the manager removing a user. The admin provides the userID to the user manager class, which updates the user class. A confirmation message is then sent to the manager.

Sequence Diagram 8: View Error Log



This is a sequence diagram of the IT admin viewing error logs for the whole system. When the IT admin requests the error logs, this request is sent to the CSV manager. The error logs is returned to the CLI controller for the IT admin to view, and a confirmation message is sent to the IT admin.

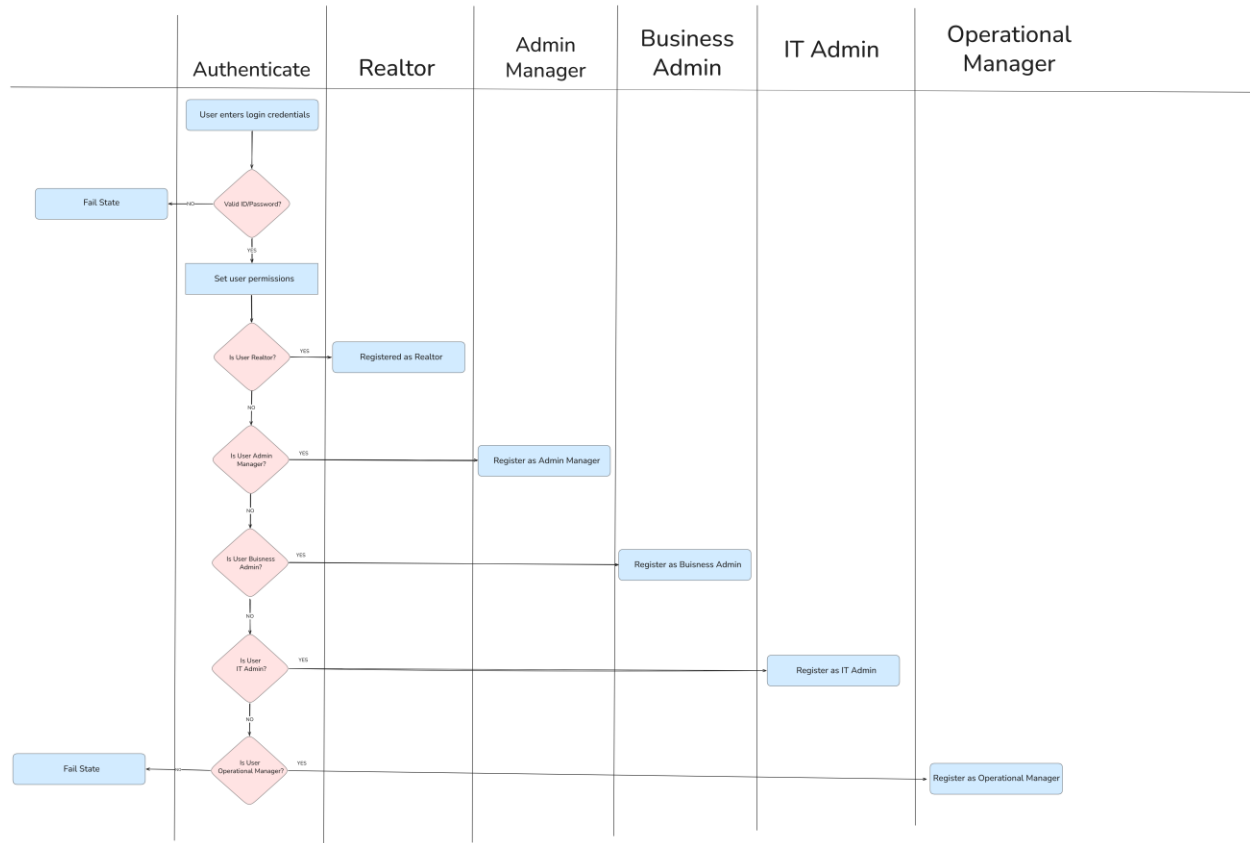
10.2 Class Diagrams



Here is a class diagram for the entire system. It shows the user class and all of its subclasses, being the realtor, operations manager, the IT admin, and the potential buyer. It also shows three other classes; being the property class, the CSV dataset, and the Docker container. The realtor adds the property class and executes the CSV dataset, the operations manager views the CSV dataset, the IT admin monitors the CSV dataset, and the Docker container containerizes the CSV dataset.

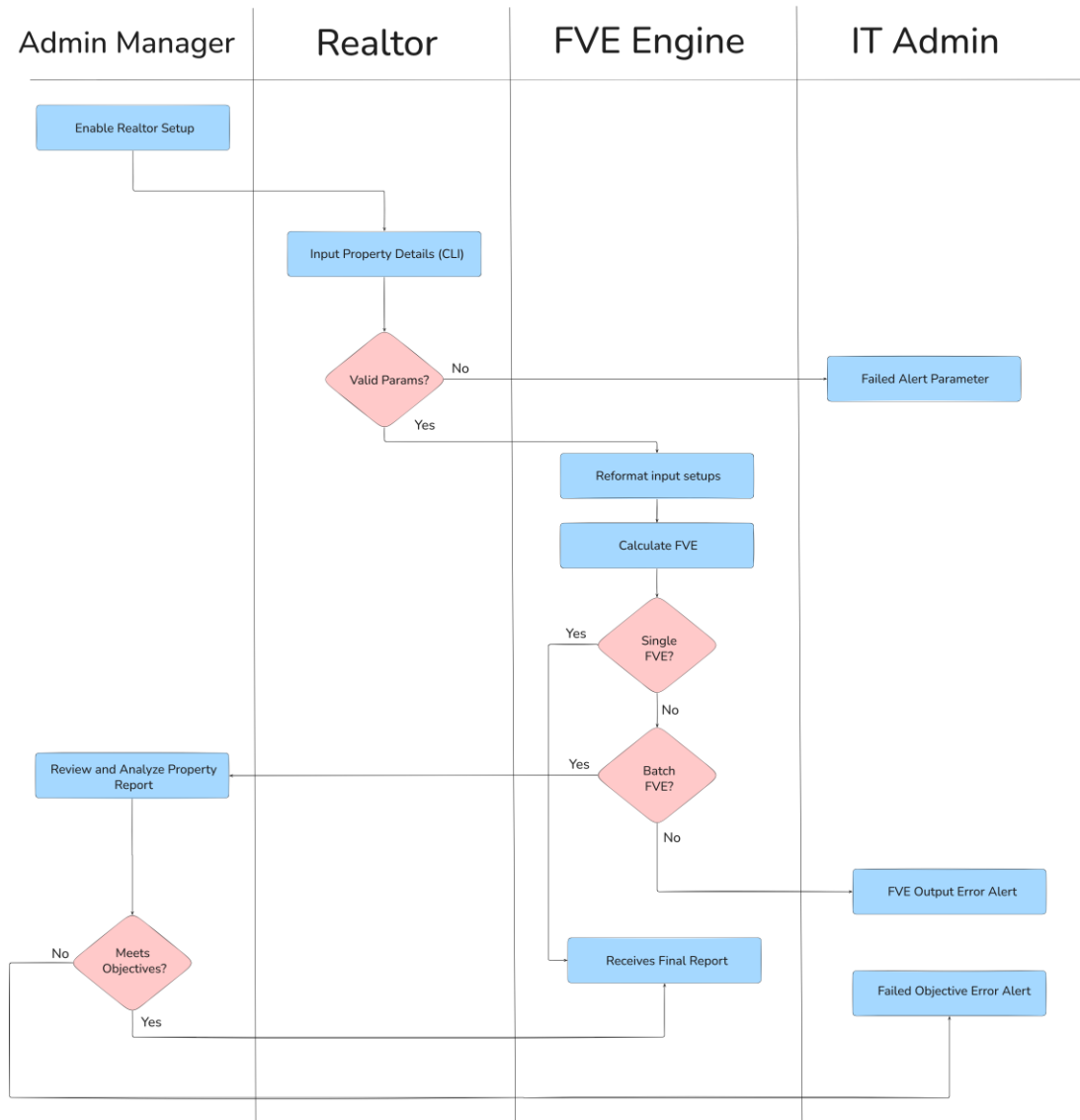
10.3 Swimlane Diagrams

Log In System



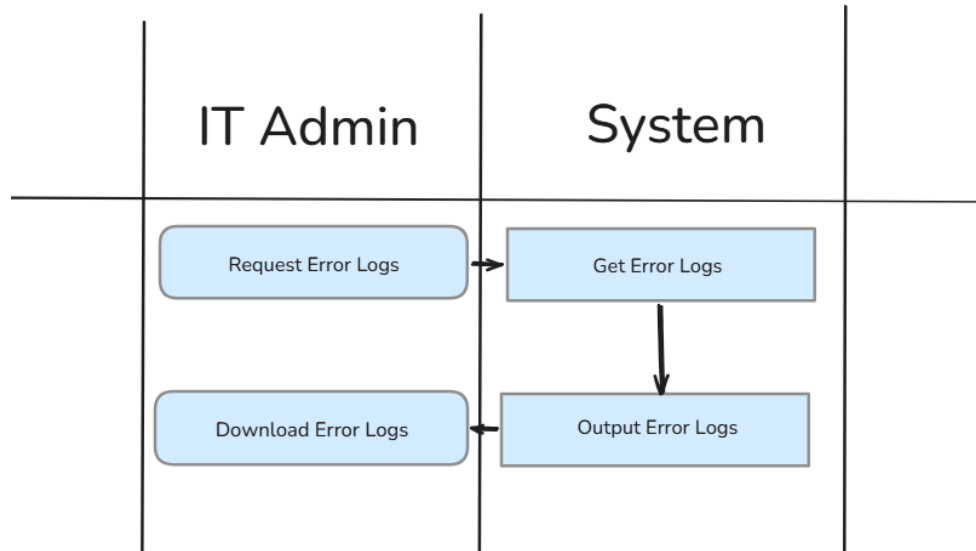
This is a swimlane diagram for the log in system. The authentication system first checks if the credentials are valid, then identifies the user's role before registering the user in the system.

FVE Engine System



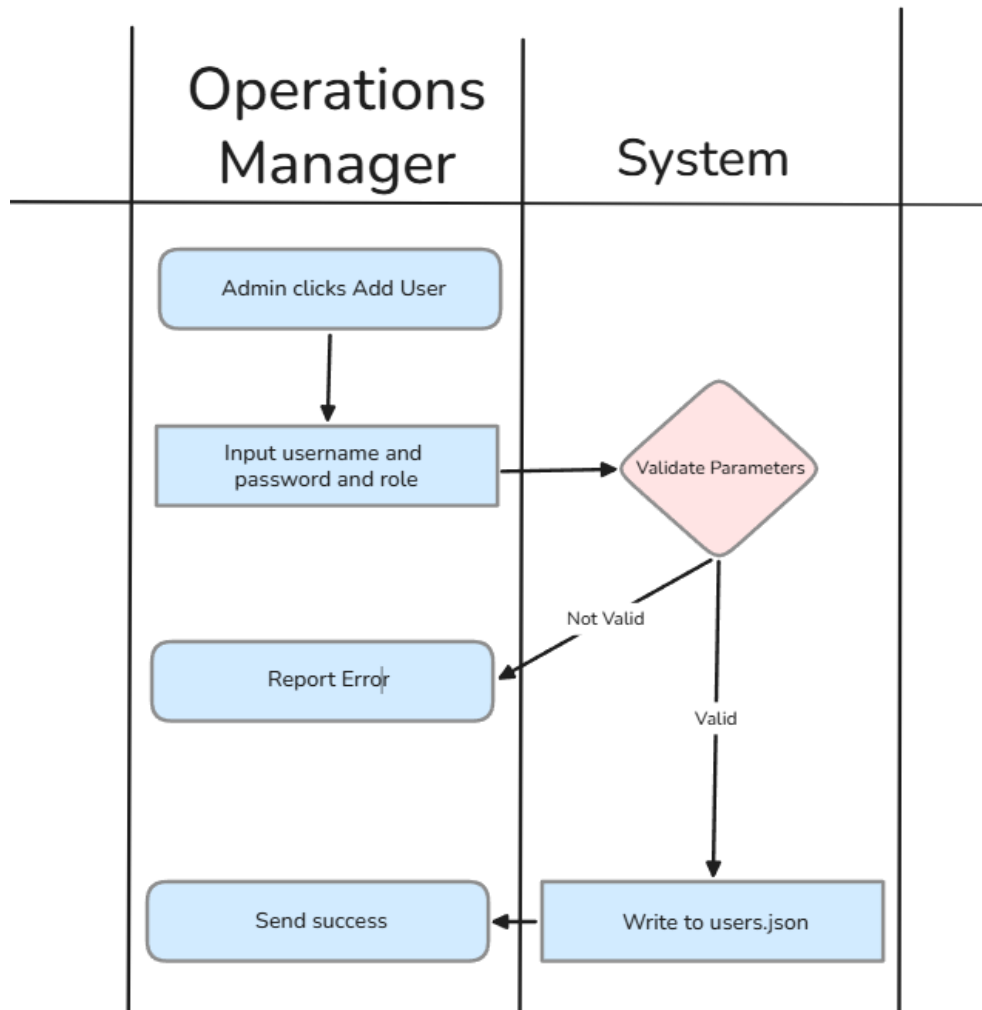
Here is a swimlane for the FVE engine. After the realtor is signed in, the realtor inputs a property dataset CSV, which is checked for valid parameters. This is brought to the FVE engine, which parses the parameters. An FVE is then calculated, after which the system checks if it should be a single FVE or a batch. A batch FVE is sent to the admin manager to review before it is given back to the FVE for the final report. A single FVE is sent straight to the final report.

Request Error Logs



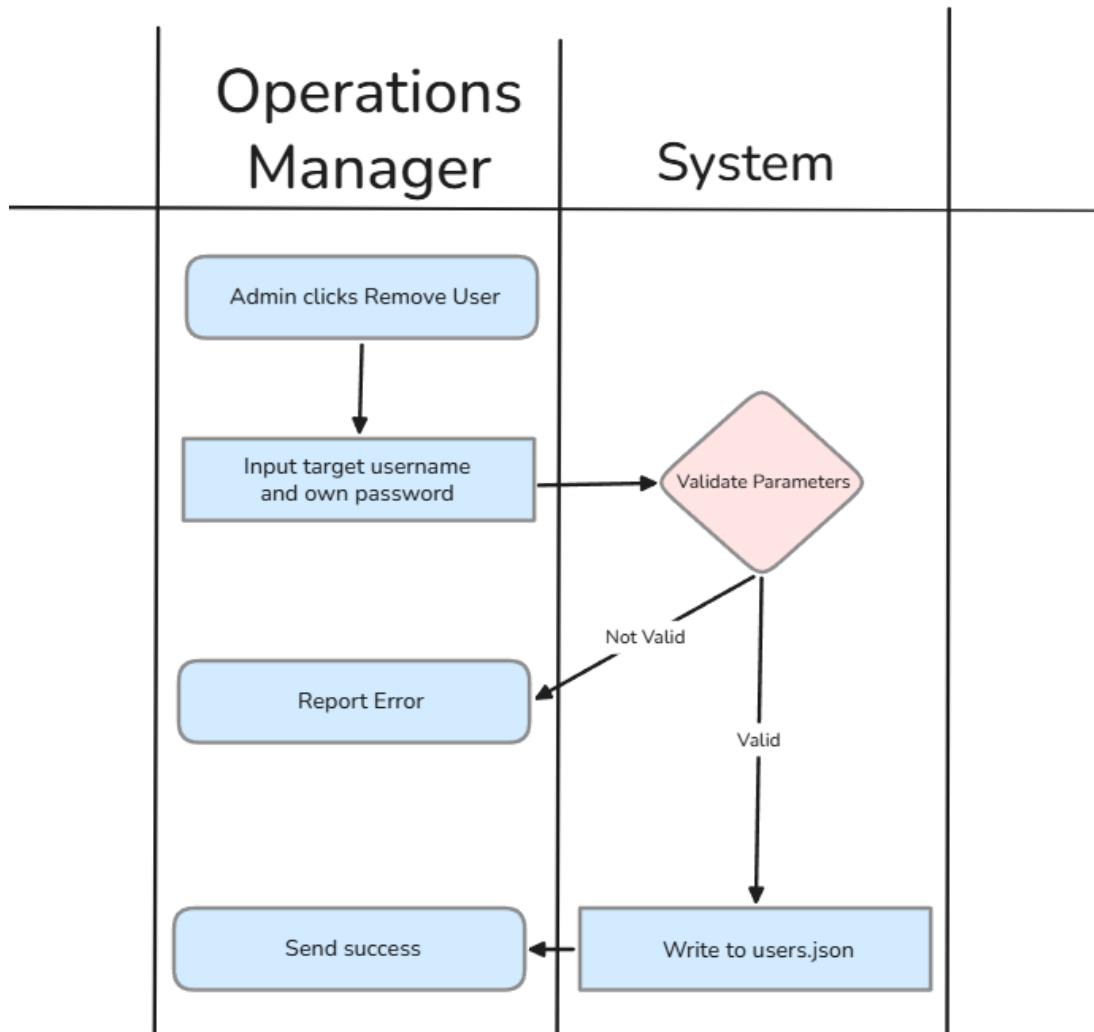
The error logs swimlane is fairly simple. The IT admin requests the error logs, and the system returns a CSV of the error logs.

Add Users



For the add users swimlane, the operations manager first inputs the username, password, and role of the user to be added. The system verifies that the parameters are correct, then writes to the users.json file the new user.

Remove Users

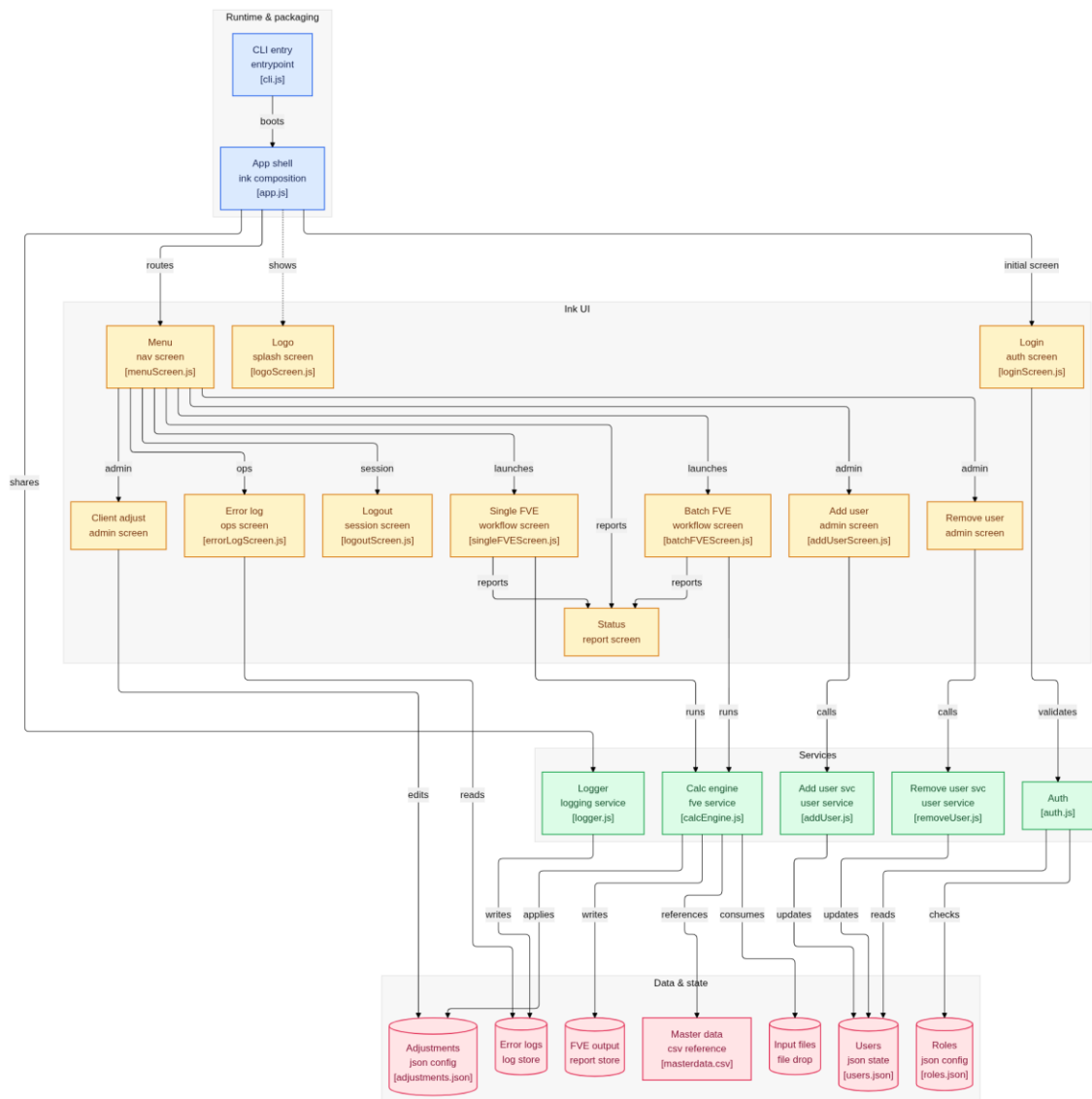


For the remove users swimlane, the operations manager first inputs the username of the user to be added and their own password. The system verifies that the parameters are correct, then writes to the users.json file to remove the target user.

11. Software Implementation

11.1 Architectural Overview

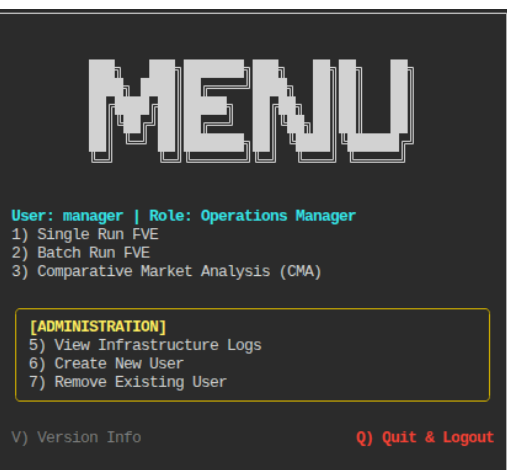
The application utilizes a modular, layered logical architecture centered around a Node.js runtime and Ink-based CLI UI composition. The system implements a secure Authentication layer and Role-Based Access Control (RBAC) which gates access to functional modules. These UI components interface with a decoupled service layer which contains the calculation matrix, user management, and authentication services. The entire system is then mounted on a File-based Persistence Layer using CSV and JSON formats. This containerized solution is deployed via Docker on a mesh network using ttyd (Web-Based Terminal), allowing authorized users to access the CLI interface directly through their web browser.



11.2 Role-Based Access Control & Authentication

The application implements a robust RBAC framework to enforce security and ensure that system functions are aligned with specific business orientations. Access is partitioned into four primary user classes: IT Admin, Operational Manager, Realtor Agent, and Business Analyst. These roles are defined within a centralized `roles.json` configuration file. User credentials are maintained within the persistence layer which the system queries during the login process for verification and authentication.

Upon successful session login, the application dynamically renders the main navigation menu which grants core functions based on the assigned role configuration.



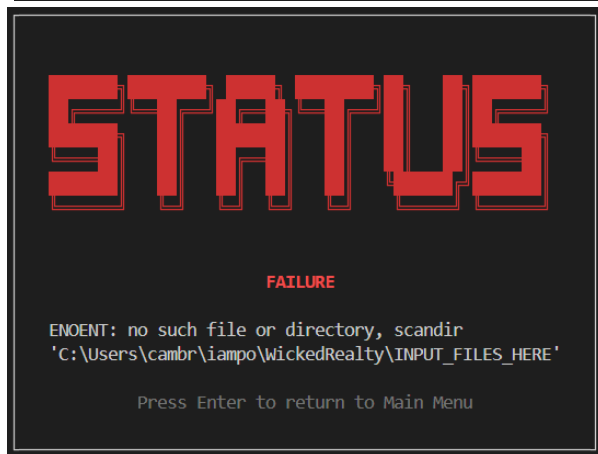
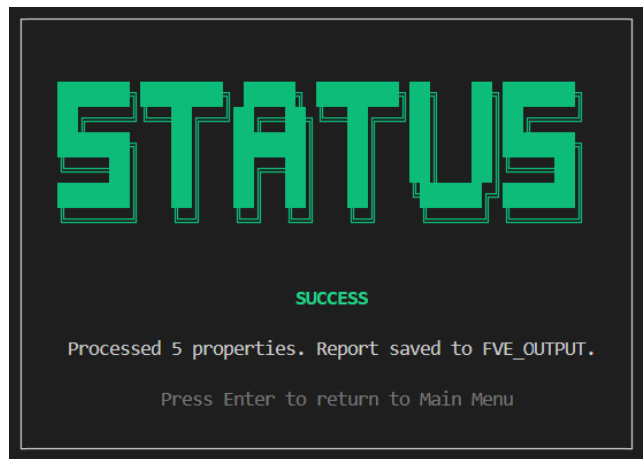


11.3 Fair Value Estimator Engine

User classes like Real Estate Agents, Business Analysts, and Operational Managers are granted access to the FVE engine to get a FVE report on either a single property or a batch of properties. This module automates property valuations through a structure pipeline of data ingestions, geospatial analysis, and comparative market analysis modelling. By evaluating similar properties against the subject “evaluatee”, the calculation engine generates THREE (3) critical datapoints: a Fair Value Estimate of the property in dollars, a count of similar listings identified and used in the generation of the FVE, and a confidence rating regarding the accuracy of the FVE.



Users are also provided with immediate visual feedback via a dedicated Status Screen where successful runs are highlighted in green with finalized reports automatically exported to the `~/WickedRealty/FVE\_OUTPUT` folder while failed runs are rendered in bold red with specific diagnostic error messaging.



The current system has been stress tested with a dataset of 20,000 listings, and it was able to generate the single FVE in under 0.2 seconds, well under the 5 second success metric that was agreed upon by the client.

11.4 System Management

The System Management module provides privileged users like IT Admin and Operational Managers with the tools necessary to maintain the application's integrity and manage user lifecycles. These administrative functions ensure that access remains secure, and that system health is monitored in real-time with ease.

This module allows authorized users to create new accounts in the system with a unique username, secure password, and user class. Conversely, the module also grants the authority to remove existing users.

USER ADD

Username: newuser
Password: ***

› **Select Role:**

- > Realtor Agent (RE)
- Operations Manager (MN)
- IT Admin (IT)
- Business Analyst (BA)

TAB: Switch Fields | ESC: Cancel

USER REMOVE

User to Remove: newuser
Your Admin Pass: ***

TAB: Switch | ENTER: Confirm Removal | ESC: Cancel

Roles with access to this module are also granted exclusive access to the Infrastructure Error Logs which aggregate recent system faults and error traces, divided by day. This provides detailed information which will allow system administrators to identify system anomalies and maintain the system throughout its lifecycle.

ERROR LOG

RECENT INFRASTRUCTURE ERRORS

```
at process.processTicksAndRejections
(node:internal/process/task_queues:81:21)]
at emitReadable_ (node:internal/streams/readable:834:12)]
at ReadStream.emit (node:events:524:28)]
at ReadStream.handleReadable
(file:///app/node_modules/ink/build/components/App.js:137:40)]
at EventEmitter.emit (node:events:524:28)]
at EventEmitter.handleData
(file:///app/node_modules/ink/build/hooks/use-input.js:84:28)]
at exports.batchedUpdates (/app/node_modules/react-reconciler/c s
/react-reconciler.production.min.js:219:357)]
at file:///app/node_modules/ink/build/hooks/use-input.js:85:21]
at file:///app/dist/screens/menuScreen.js:23:15]
[2026-05-01T18:07:5 [UI_STRESS_TEST] Error: MANUAL_TEST: User
713Z] triggered a synthetic system failure.
```

Press R to Refresh | Press ENTER for Status | Press ESC for Menu



The application was containerized in Docker and deployed on a mesh network which allows authorized users to freely connect via the assigned port from the convenience of their web browser.

12. AI Usage Reflection

12.1 RD1.0 AI Usage

Our team originally tried using our preferred LLM models, e.g. SurfSense/Gemini, to give us a few ideas about how to fill each section. However, when attempting to prompt, we were constantly recommending an obscene amount of microservices that we didn't exactly need given our project was for a small-scale modern application to solve a niche need.

While we attempt to guide the LLMs toward using less microservices/smaller infrastructure, it took much more refining and hand holding than we wanted to do. Hence, we ended up writing all the document ourselves given: a) we're implementing this product, so knowing what is being put down is more useful than having it generated, b) any issues/questions are more quickly addressed during this planning phase, c) each group member knows their part more clearly and roles are easier to distribute.

Examples of us guiding the LLMs included using words such as "hypothetically", "no microservices", "small scale" and directly feeding resources that we desired it to emulate (such as the proposal document). While it generated *ok* answers, there was a lot left to be desired.

The main software aid we used in writing was a spellcheck which MS word provided by default. However, given this isn't generative (as there isn't exactly any prompting or tokens being used), we don't really have much else to provide for this section. Grammarly was used for some sections, but again, was used for the spell-checking features, not for any generative usages (again no prompts or anything used here).

12.2 RD2.0 AI Usage

Multiple AI aids were used in the creation of RD2.0. To start, ChatGPT was used to generate the equations in the revisions in section 2.d. The equations were then edited to better fit the requirements. The conversation can be seen here <https://chatgpt.com/share/69d31ef6-5648-832e-b6f5-f501e3099fd5>.

The next use of AI was Gemini 3.1. We were unsure how backwards traceability and forwards traceability should look in section 5, so we generated templates for both using Gemini 3.1. This saved time as we were able to get a sense of direction on what each section should look like, which lets us fill in with our own information. Section 5.2-5.3 used ChatGPT to help Jax look through our overall document and find the correct links for forward and backwards traceability. Section 5.6 was explored with ChatGPT as there were multiple ways to store the output, and we wanted to view all our options. Ultimately, we decided to output a CSV. The conversation can be seen here <https://chatgpt.com/share/69d322f8-0c18-8325-9ded-d7c6a718542b>.

<https://chatgpt.com/share/69d32b34-5088-8329-be16-b2c571a83460>

On section 6.1, Gemini 3.1 was used to suggest ideas on what UI external interface requirements were useful. AI suggested a color differentiated text coloring to improve readability between successful executions, non-fatal warnings and critical errors. This can be further explained below.

Section 7 uses Claude to revise some requirements for reliability, availability, and scalability. For better context, an imported R.D. 1.0 was added to get a foundation of our project goals and ideas. Added to the prompt was the feedback that the TA has given us members, ensuring that it gives us better detailed output for what necessary steps for each requirement. Claude AI suggested adding various requirements that resigned with the project to reliability, availability, and its scalability. We also provided constructive feedback about our own ideas for specific required sections and asked if that resided well with the project, resulting in a more defined and clear explanation that is more desirable to the detail we needed.

<https://claude.ai/share/38507f24-ae9f-4378-b34e-503cf4bf5366>

Section 8 used Gemini 3.1 to help find directions in the slides for each diagram type mentioned. Gemini was then asked to help decide on an architecture type, eventually landing on a layered approach and to provide a picture, which was discarded. From there, further questions were

asked to clarify directions for some diagrams, like the use-case diagram. This discussion can be found here <https://gemini.google.com/share/842fdc286a0a>.

Ephraim

Section 5-4

Gemini 3.1 Prompt: Give me examples on how forward traceability can be integrated into section 5-4 of the document.

Response: “**Verification:** User Acceptance Testing (UAT) where agents or system analysts trigger the CMA report generation for a target property via the platform interface. Testers will download the resulting CSV file and open it in a standard spreadsheet application (e.g., Excel or Google Sheets) to verify that the file structure is intact (no broken columns), and that the computed FVE, geospatial parameters, and list of comparative properties are accurately mapped to their respective columns. Automated integration tests will also verify that the CSV file is generated within the expected time limit and correctly saved to the designated file directory.”

How it was used: Section 5-4 adopted the User Acceptance Testing suggestion whereby testers and agents can test as they use the system. The automated integration tests were also adopted to verify the functionality of the system feature.

Gemini 3.1 Prompt: Show me an example of a full user story for IT administrator including narrative and acceptance statements.

Response: “**Narrative:**

- **As an** IT Administrator,
- **I want to** access detailed error logs whenever the FVE calculation or CSV generation process fails,
- **so that** I can rapidly troubleshoot API timeouts, missing geospatial data, or file storage permission errors without relying solely on agent bug reports.

Acceptance:

- **Given** the FVE engine encounters a system error (e.g., a File Write/Permission Failure during CSV export),
- **When** the system halts the process and returns an error to the user,

- **Then** the system writes a standardized log entry to the administrative console containing the timestamp, specific error code, and the property ID that triggered the failure.

”

How it was used: This template gave us a good idea over how to structure the user stories section and was helpful in understanding how an acceptance statement was written. We adopted this template for Section 5.

Section 6.1

Gemini 3.1 Prompt: Give me examples what are some user interface requirements that should included.

Response: “

a. User Interfaces This section describes the boundary between the human user and the system. Given the constraints of the project, the primary interface is strictly text-based and accessed via a standard terminal emulator (e.g., Windows Command Prompt, PowerShell, macOS Terminal, or Linux bash/zsh).

- **Format & Style:** * The interface shall be a Command Line Interface (CLI) designed for minimalism, speed, and low resource overhead.
 - The system shall utilize standard output (stdout) for successful executions and standard error (stderr) for failure states.
 - To improve readability for non-technical real estate agents, the CLI shall implement basic text coloring (e.g., Red for errors/exceptions, Green for successful valuations/exports, and Yellow for warnings like missing optional data).
- **Interaction Constraints:**
 - **Command Execution:** Users will interact with the system using standard POSIX-style command syntax consisting of a base command, an action, and specific flags/arguments.
 - *Example Syntax:* fve-engine calculate-solo --file <property_filename.csv>
 - *Example Syntax:* fve-engine calculate-batch --dir <directory_path> --radius 5
 - **ASCII Menu Navigation:** For users uncomfortable with raw command syntax, executing the base command (e.g., fve-engine) without arguments will trigger an interactive ASCII-formatted text menu. This menu will allow users to select

operations (e.g., "1. Run Single FVE", "2. Run Batch CMA") using numeric key presses.

- **Feedback Mechanisms:** For batch processing (up to 100 properties), the interface shall display an ASCII progress bar or percentage indicator to provide the user with real-time feedback on system execution.
- **Help Dialogues:** Appending --help to any command will output a formatted table of valid syntax, required arguments, and usage examples directly into the terminal window.

How it was used: I really liked how basic requirements like color coded texts could make a really large impact in terms of bridging the gulf of execution and evaluation. We adopted the idea that the CLI would STDOUT green for successes, yellow for warning, and red for stopping failures. This is often overlooked and not part of most CLI applications, which I personally find wanting.

12.3 Arika's AI Usage Stack

Section 6.2 onwards was all written by hand with NO AI involved. Additionally, much of Section 7 was overhauled by hand as a lot of written prompts did not make any sense. They were written to confirm with the docker standards outlined as written by hand in section 6.2 onwards. Many sections were modified in order to cleanly flow from section to next section. This was done by hand without AI.

The overall knitted document used a variety of AI agents to double-check the entire document. The workflow is as follows:

DISCLAIMER: As with all AI tools, I understand that they are good for generating templates not for actual code. Overly relying on AI generated code quickly leads to immense tech debt and what many people call "vibe coding clean-up" roles. Most of what was used was either a) proof of concept to see how well the tools fair with heavy user influence, or b) to check existing work currently written as a "helper" to optimize performance and readability.

While documentation was tidied and used to generate suggestions with the Gemni CLI workflow, most of it still ended up handwritten.

The primary use of AI used two main tech stacks:

- surf sense (<https://www.surfsense.com/>)
- gemni cli (<https://github.com/google-gemini/gemini-cli>)

GSD (getshitdone) was used for the initial project planning (as seen below). Then I manually went in and edited by hand parts that didn't seem good, or fit with the code structure.

Gemni CLI

I used the following GSD model ontop of gemni CLI to better verify the work and assumptions done:

- getshitdone (<https://github.com/rokiCool/gsd-opencode>)

Additionally, I have my own custom rule-set script which is pretty long so won't be included here.

Several model types were used dependent on what needed to be done.

As with most GSD workflows for each new PDN project I would go with the following:

- Initialization (/gsd-new-project)
 - Initial questions
 - PDF Document upload of problems
 - Assumptions being made (and fixed)
 - Code style wished to be generated
 - codebase linked
 - README generated
 - Point towards internal data that should be used (all 3 textbooks, path to local repo)
 - Research
 - Custom phase with gemni
 - able to do more in depth analysis of the project given along with sources pointed to
 - additionally looks out for more sources online
 - I usually search myself for stackoverflow topics I may think useful. I will then point it to my surfsense model to see if it makes sense to implement it, then feed it to GSD to write in.
- Planning (/gsd-plan-phase)
 - Split each problem and sub problem into their own atomic phases/tasks
 - A plan checker ensuring nothing was missed
 - Context Isolation and verification of tasks
 - Execution (/gsd-run-phase)
 - execution of per phase as outlined in generated README
 - tasks are run per wave with atomic commits

- Each commit is then showed of what changed and I can go in and ask / edit the changes it made

- Verification

- Spawn any debug agents

- generate potential ticket issues

How Agents Work in Gemini CLI

In the Gemini CLI and GSD workflow, "agents" are specialized AI instances that can be spawned to handle specific, focused tasks. Think of them as delegating a piece of your project to an

expert who specializes in that one area. This is a core concept for efficiently managing complex software projects.

Key Concepts:

- Delegation: Instead of having one single AI context trying to manage everything from high-level planning to low-level code fixes, you delegate. For example, when you run `/gsd:plan-phase`, you are spawning a `gsd-planner` agent whose sole job is to create a high-quality, executable plan.
- Specialization: Each agent is optimized for its role.
 - `gsd_phase_researcher`: Scours the web and provided documents for information.
 - `gsd-planner`: Breaks down goals into concrete, verifiable steps.
 - `gsd_codebase_mapper`: Analyzes existing code to understand its structure and conventions.
 - `gsd_verifier`: Checks if the completed work actually meets the original goals.
 - `generalist`: A flexible agent that can be given a wide range of tasks, often used for batch operations or speculative investigations without cluttering the main chat.
- Parallelism & Isolation: Agents can often work in parallel. You could have one agent researching an API while another maps out the existing codebase, speeding up the overall process.

Their work is also isolated, preventing them from interfering with each other.

- Workflow Orchestration: The main GSD process orchestrates these agents. It spawns them as needed, feeds them the necessary context, and integrates their results back into the main project plan or codebase. For instance, the output of the `gsd-phase-researcher` (a `RESEARCH.md` file) becomes the input for the `gsd-planner`.

By breaking down a large, ambiguous goal into a series of smaller, well-defined tasks handled by specialized agents, the GSD framework provides structure, reduces errors, and ensures all aspects of the initial request are met. You, the user, can oversee this process, verifying the output of each agent at each stage.

Surf Sense

On top of the current sourcing, surf sense was used more as a generalized explainer rather than giving answers for robust problems. Surf Sense was hosted on a docker container, with an exposed port on my home server, using tailscale to connect to the endpoint. Several connectors were used to gather information from the following sources

- Obsidian
- Unraid
- outlook
- google notebook
- My github

These sources are then used during query and write time to generate quick answers. Surf Sense is generally used as a search feature to find any pre-existing sources I may have on a topic in order to help gemni+GSD run better.

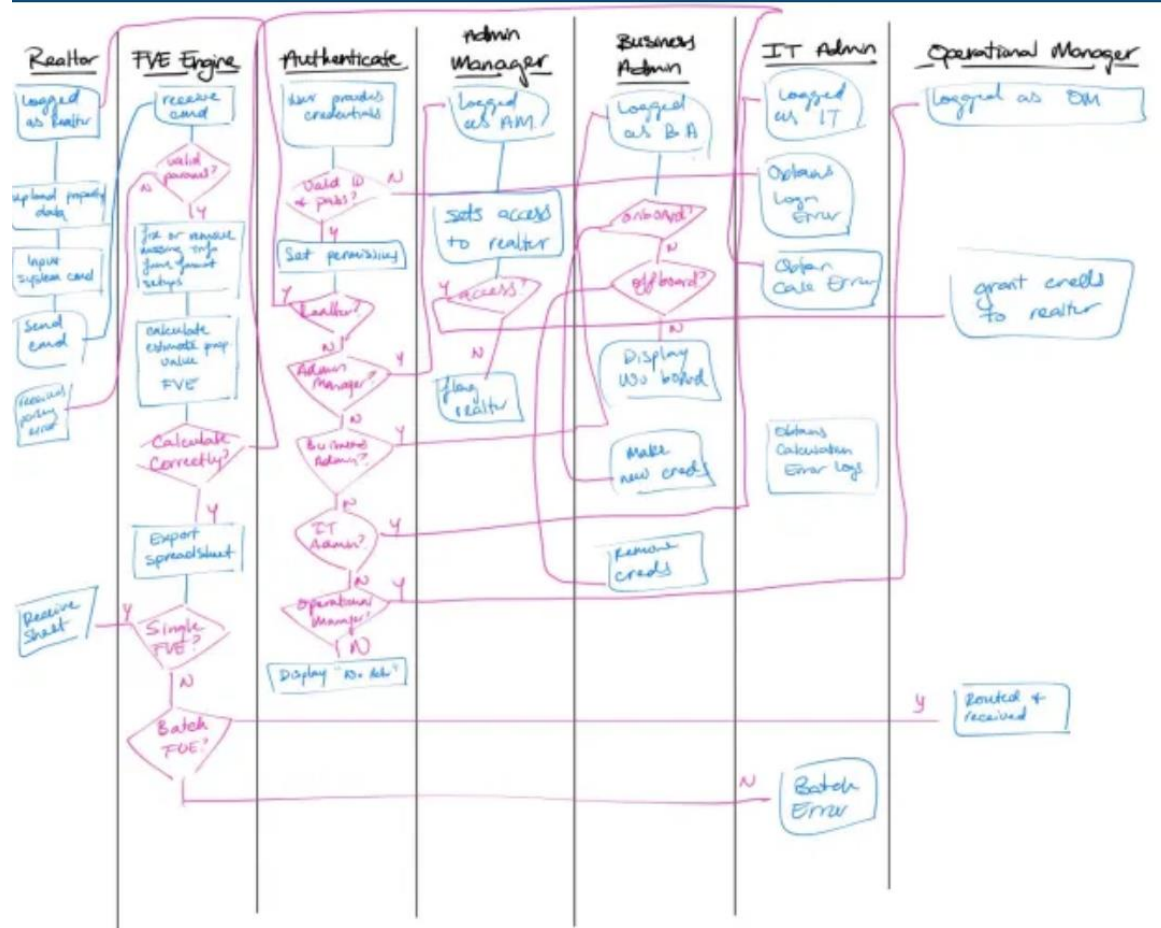
12.4 RD 3.0 AI Usage

RD 3.0 was created with minimal AI usage. AI was only used for editing RD 2.0 based off TA feedback and for the creation of the FVE engine itself. Individuals who used AI and performed these tasks go into their usage in detail below. For the creation of the engine, Gemini was given an overview of the purpose, its architecture, and its tech stack before any specific questions were asked. Any other section of the RD, presentation, or application not mentioned below was made in its entirety without AI.

Thuan's AI Usage:

For my AI Usage, I aimed to process improvements with the feedback from the TA back on RD 2.0. Using AI, I summarized each RD file for context to provide key areas in which parts were needed to improve (<https://gemini.google.com/share/36247eaf84e7>), including unrealistic criteria numbers and diagram formatting. For the swimlane diagrams, no AI has been utilized, as most of the workflow was more so generalized based on the format of which various visualized actors are shown. The image provided below showcases a sketch draft of the

swimlane diagram in which the project was operated:



Ephraim's AI Usage

Prompt: Does Ink js have a screen handler or a router similar to React Router which can dynamically and conditionally render screens in the application based on userState?

Gemini 3 Response:

Ink does not have a "built-in" screen handler or router component in its core library. Instead, because Ink is a standard React renderer, it relies on the React ecosystem and standard React patterns to handle navigation.

According to the documentation and the repository structure, there are two main ways to handle multiple screens:

1. Official Recommendation: MemoryRouter

The Ink repository includes a specific "**Routing Recipe**" that suggests using **React Router** with a **MemoryRouter**. Since there is no browser history in a terminal, **MemoryRouter** manages the navigation state in memory.

- **How it works:** You define your routes just like a web app.
- **Location in Docs:** You can find this in the `recipes/` folder of the GitHub repo.

JavaScript

```
import {MemoryRouter, Routes, Route} from 'react-router-dom';

const App = () => (
  <MemoryRouter>
    <Routes>
      <Route path="/" element={<LogoScreen />} />
      <Route path="/login" element={<LoginScreen />} />
    </Routes>
  </MemoryRouter>
);
```

2. Manual State Management (The "Standard" Way)

The most common pattern used by the Ink community (and often seen in their examples) is simple conditional rendering using **React useState**. This is the method we've been using in your project so far. It is lightweight and doesn't require extra dependencies like **react-router**.

3. Community Libraries

If you want something more specialized for the terminal, there are community-made hooks and components, though they aren't part of the core ink package:

- **ink-router:** A dedicated router built specifically for Ink.
- **ink-use-router:** A hook-based approach for simple navigation.

Summary

If your app (Wicked Realty) is going to have 10+ screens with complex "back" button logic, following the **React Router (MemoryRouter)** recipe is the most robust path. If you only have a few screens (Logo -> Login -> Menu), the **conditional state** approach is much faster to implement and easier to debug.

How it was used: I opted to use the "standard router" approach and implement manual routing.

Prompt: What libraries should I consider to implement a loading spinner?

Gemini 3:

Yes, you can integrate a spinner very easily. In the Ink ecosystem, there is a dedicated component called **ink-spinner** that handles the animation timing for you.

1. Install the Spinner

Run this in your project directory:

```
Bash
npm install ink-spinner
```

2. Integrate it into your Status Screen

You can replace the " ⌚ " emoji with the animated component. The spinner will automatically stop animating when its parent component (the "loading" block) is unmounted or hidden by your state change.

Update source/screens/ReportStatusScreen.js:

```
JavaScript
import React, { useState, useEffect } from 'react';
import { Box, Text } from 'ink';
import Spinner from 'ink-spinner'; // Import the spinner

const ReportStatusScreen = ({ onBack }) => {
```

```

const [status, setStatus] = useState('loading');
const [message, setMessage] = useState('Processing CSV datasets...');

// ... (Your useEffect logic from before) ...

return (
  <Box flexDirection="column" borderStyle="round" padding={1} width={50}>
    <Text bold>CMA Generation Pipeline</Text>

    <Box marginTop={1}>
      {status === 'loading' && (
        <Box>
          <Text color="yellow">
            <Spinner type="dots" /> { /* This adds the animation */ }
          </Text>
          <Text> {message}</Text>
        </Box>
      )}

      {status === 'success' && <Text color="green">✓ {message}</Text>}
      {status === 'error' && <Text color="red">✗ {message}</Text>}
      {status === 'warning' && <Text color="blue">⚠ {message}</Text>}
    </Box>
  </Box>
);
};

```

3. Choosing Your Style

ink-spinner uses **cli-spinners** under the hood. You can change the type prop to match the "Wicked Realty" aesthetic:

- dots (Default)
- line (Classic / - \ |)
- earth (A rotating globe)
- moon (Moon phases)

- bouncingBar

Pro-Tip: Why the Spinner works so well in Ink

In a traditional CLI, you have to manually clear the line and re-print the spinner frames using intervals. In Ink, because it uses the **React Reconciler**, the spinner component simply triggers a re-render of its own small section of the terminal every ~80ms. This keeps your CPU usage low while making the app feel "alive" during heavy CSV processing.

One quick check for your environment:

Since you are using **Ubuntu**, most standard terminal emulators (like GNOME Terminal or Tilix) support the Unicode characters used by the dots spinner. However, if you see weird blocks instead of dots, try switching the type to line.

Are you planning to add a progress bar as well if the CSV has a lot of rows, or is the spinner enough for now?